

ALGEBRAIC HOLOGRAPHIC HT SECTOR WITH VIRASORO BOUNDARY AND DERIVED CHIRAL CENTRE BULK

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ABSTRACT. Three-dimensional Einstein gravity with negative cosmological constant is $SL(2, \mathbb{R}) \times SL(2, \mathbb{R})$ Chern–Simons (Achúcarro–Townsend, Witten); its boundary symmetry algebra is the Virasoro algebra at central charge $c = 3\ell/(2G)$ (Brown–Henneaux). We identify the *algebraic holographic HT sector* of this theory: the open Beilinson tower with boundary $\partial = \text{Vir}_c$, bulk $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) = C_{\text{ch}}^{\bullet}(\text{Vir}_c, \text{Vir}_c)$, and interaction the $\text{SC}^{\text{ch}, \text{top}}$ -brace structure on the pair $(Z_{\text{ch}}^{\text{der}}(\text{Vir}_c), \text{Vir}_c)$. The ordered bar complex $\tilde{B}^{\text{ord}}(\text{Vir}_c)$ on $\mathbb{C} \times \mathbb{R}$ is the twisting coalgebra; the derived chiral centre carries E_3^{top} structure via Sugawara topologisation (cohomological + weight-completed scope); the universal Maurer–Cartan element Θ_{Vir_c} generates the shadow obstruction tower $\{S_r\}_{r \geq 2}$ recording the boundary-to-scalar trace of the algebraic sector.

The trace direction (operators $\rightarrow$ scalar) is well-defined and unconditional on the Virasoro Koszul locus: $\kappa(\text{Vir}_c) = c/2$ is the Brown–Henneaux central charge; the quartic shadow $10/[c(5c+22)]$ matches the level-4 Kac determinant; the shadow free energies $F_g = \kappa \lambda_g^{\text{FP}}$ reproduce the perturbative one-loop Hodge-integral expansion. The inverse direction (scalar $\rightarrow$ operator), in particular the construction of a compact dynamical-metric path integral from the algebraic sector, is the Vol I open frontier F12: it requires the BTZ / Cardy hypothesis package (modular invariance, vacuum dominance, saddle stability) together with the Maloney–Witten plus Manschot–Moore Farey-tail modular completion. Koszul duality $\text{Vir}_c \leftrightarrow \text{Vir}_{26-c}$ is the algebraic shadow of S-duality; the self-dual point sits at $c = 13$. The algebraic Page curve emerges from complementarity: $\kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = 13$ forces saddle exchange at $t_P = 3S_{\text{BH}}/13$, c -independent. The crossing is the structural shadow of the Page transition; the full information-theoretic content remains controlled by the island mechanism (Penington, AEMM). BTZ black holes are Maurer–Cartan elements of the gravitational convolution algebra; de Sitter entropy arises by analytic continuation $\kappa \rightarrow -|\kappa|$.

CONTENTS

I. THREE-DIMENSIONAL GRAVITY AS CHERN–SIMONS

Three-dimensional Einstein gravity with negative cosmological constant $\Lambda = -1/\ell^2$ has no propagating degrees of freedom. The Riemann tensor is determined algebraically by the Ricci tensor, which equals $\Lambda g_{\mu\nu}$; every solution is locally AdS_3 . The local triviality has an exact formulation: pure gravity in 2+1 dimensions is a gauge theory.

I.1. The Chern–Simons formulation.

Theorem I.1 (Achúcarro–Townsend [?AT86], Witten [?Wit88]). *Three-dimensional gravity with cosmological constant $\Lambda = -1/\ell^2$ is classically equivalent to $SL(2, \mathbb{R}) \times SL(2, \mathbb{R})$ Chern–Simons theory at levels $k_{\pm} = \ell/(4G)$, where G is Newton’s constant. The gauge fields are*

$$A_{\pm}^a = \omega^a \pm \frac{e^a}{\ell},$$

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where e^a is the dreibein and ω^a the spin connection. Flatness $F_{\pm} = dA_{\pm} + A_{\pm} \wedge A_{\pm} = 0$ is equivalent to constant negative curvature plus torsion-freedom.

Proof. The Einstein–Hilbert action with $\Lambda < 0$ in three dimensions reads

$$I_{\text{EH}} = \frac{1}{16\pi G} \int \epsilon_{abc} \left(e^a \wedge R^{bc} + \frac{1}{3\ell^2} e^a \wedge e^b \wedge e^c \right),$$

where $R^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb}$ is the curvature two-form. The Chern–Simons functional for a gauge field A on a three-manifold is $I_{\text{CS}}[A] = (k/4\pi) \int \text{tr}(A \wedge dA + \frac{2}{3} A \wedge A \wedge A)$. Using the $\mathfrak{sl}_2$ trace $\text{tr}(J_a J_b) = \frac{1}{2} \eta_{ab}$ and the decomposition (??):

$$I_{\text{CS}}[A_+] - I_{\text{CS}}[A_-] = \frac{k}{4\pi} \int \epsilon_{abc} \left(2 e^a \wedge R^{bc} + \frac{2}{3\ell^2} e^a \wedge e^b \wedge e^c \right) = \frac{k}{2\pi} I_{\text{EH}},$$

provided $k = \ell/(4G)$. The equations of motion $F_{\pm} = 0$ decompose into the torsion constraint $T^a = de^a + \epsilon^a_{bc} \omega^b \wedge e^c = 0$ and the Einstein equation $R^{ab} + (1/\ell^2) e^a \wedge e^b = 0$. $\square$

Remark 1.2 (Scope: algebraic sector versus dynamical-metric path integral). The Achúcarro–Townsend–Witten equivalence holds at the classical level: both theories have the same equations of motion and the same moduli of flat connections. At the quantum level, the Chern–Simons path integral is a topological invariant (the Reshetikhin–Turaev functor), while a compact dynamical-metric gravitational path integral must also sum over topologies and saddle geometries.

The *algebraic holographic HT sector* of the gravitational theory is the open Beilinson tower $\partial = \text{Vir}_c$, bulk $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) = C_{\text{ch}}^{\bullet}(\text{Vir}_c, \text{Vir}_c)$, interaction the $\text{SC}_{\text{ch}}^{\text{ch}, \text{top}}$ -brace structure on $(Z_{\text{ch}}^{\text{der}}(\text{Vir}_c), \text{Vir}_c)$. This sector is the level-3 / level-4 / level-5 Beilinson rungs of the gravitational theory; it is unconditional on the Virasoro Koszul locus (generic c).

The *trace direction* (operators $\rightarrow$ scalar) descends from the algebraic sector to modular and partition-function shadows by trace + clutching on the open category; this is the content of Sections ??–?? and is unconditional. The *inverse direction* (scalar $\rightarrow$ operator), in particular the construction of a compact dynamical-metric path integral from the algebraic sector, is the Vol I open frontier F12 and requires a distinct hypothesis package: the BTZ / Cardy hypotheses (modular invariance + vacuum dominance + saddle stability) together with the Maloney–Witten plus Manschot–Moore Farey-tail modular completion. The reverse direction is a separate conjectural inverse problem, stated in Section ?? and Section ??.

1.2. The Brown–Henneaux central charge. The boundary dynamics is not trivial. Imposing asymptotically AdS_3 boundary conditions at the conformal boundary $\partial(\text{AdS}_3) \cong \mathbb{R} \times S^1$ produces a two-dimensional conformal field theory whose symmetry algebra is generated by the asymptotic charges.

Theorem 1.3 (Brown–Henneaux [?BH86]). *The asymptotic symmetry algebra of AdS_3 gravity with $\Lambda = -1/\ell^2$ is two copies of the Virasoro algebra at central charge*

$$c = \frac{3\ell}{2G},$$

equivalently $c = 6k$ with $k = \ell/(4G)$ the Chern–Simons level.

Proof. In the Chern–Simons formulation, the boundary condition on the half-space $\mathbb{R}_{\geq 0} \times \mathbb{C}$ is the Fefferman–Graham gauge $A_z|_{t=0} = J_- + T(z) J_+$, where $J_{\pm}, J_0$ are the $\mathfrak{sl}_2$ generators with $[J_0, J_{\pm}] = \pm J_{\pm}$, $[J_+, J_-] = 2J_0$. The residual gauge transformations preserving this form are $g = \exp(\epsilon(z) J_+)$, acting by

$$\delta_{\epsilon} T = \epsilon' T + 2\epsilon T' + \frac{c}{12} \epsilon''',$$

which is the Virasoro coadjoint action. The Chern–Simons symplectic form induces the Poisson bracket

$$\{T(z), T(w)\}_{\text{PB}} = \frac{c}{2} \delta'''(z-w) + 2T(w) \delta'(z-w) + T'(w) \delta(z-w),$$

which is the defining relation of Vir_c . The central charge is $c = 6k = 3\ell/(2G)$. $\square$

1.3. The input datum. The entire construction of this paper flows from the Virasoro λ -bracket, which encodes the OPE of the boundary stress tensor in algebraic form:

$$\{T_\lambda T\} = \partial T + 2T\lambda + \frac{c}{12} \lambda^3.$$

The three terms encode, respectively: the translation $T_{(0)}T = \partial T$ (simple pole), the dilatation $T_{(1)}T = 2T$ (double pole), and the central extension $T_{(3)}T = c/2$ (quartic pole). In the OPE formalism:

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The triple pole $T_{(2)}T = 0$ vanishes: the coefficient of λ^2 in the λ -bracket is zero, forced by skew-symmetry. The quartic pole is the single datum from which the entire gravitational structure follows.

Remark 1.4 (Physical parameters). The central charge $c = 3\ell/(2G)$ is the only free parameter of 3d gravity with $\Lambda < 0$. In Planck units $\ell = 1$, it is $c = 3/(2G)$: the inverse coupling constant. The semiclassical limit $c \rightarrow \infty$ is $G \rightarrow 0$. The curvature $\kappa(\text{Vir}_c) = c/2$ records the same information; the relation $c = 2\kappa$ persists throughout.

1.4. The holomorphic-topological structure. The Chern–Simons theory on $\mathbb{R}_{\geq 0} \times \mathbb{C}$ has a natural factorisation into holomorphic and topological directions: the $\mathbb{C}$ -direction carries the holomorphic OPE structure (the vertex algebra of the boundary), while the $\mathbb{R}_{\geq 0}$ -direction carries the topological ordering (the associative structure of the bar complex). The product $\mathbb{C} \times \mathbb{R}$ is the habitat of the E_1 -chiral bar complex $\bar{B}^{\text{ord}}(\text{Vir}_c)$, which is a dg coalgebra over the Koszul dual cooperad $(\text{ChirAss})^!$: the bar differential uses the holomorphic OPE data from $\text{FM}_k(\mathbb{C})$, while the deconcatenation coproduct uses the ordered topological structure from $\text{Conf}_k^<(\mathbb{R})$.

The $\text{SC}^{\text{ch}, \text{top}}$ structure on the chiral derived centre $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c)$ (see Section ??) becomes E_3 -topological via Sugawara topologisation at non-critical level: the Sugawara stress tensor $T(z)$ satisfies $T = \{Q_{\text{CS}}, G'\}$ in the BRST complex, so $\mathbb{C}$ -translations are Q -exact, the complex structure on $\mathbb{C}$ becomes irrelevant in cohomology, and the two $\text{SC}^{\text{ch}, \text{top}}$ colours collapse. The result is an E_3 -topological algebra: $E_2^{\text{top}} \times E_1^{\text{top}}$ via Dunn additivity, independent of the complex structure on $\mathbb{C}$. This is proved for the affine Kac–Moody algebra $V_k(\mathfrak{sl}_2)$ at non-critical level $k \neq -2$; for the Virasoro algebra obtained by Drinfeld–Sokolov reduction, the topologisation is cohomological and the chain-level structure is conditional on A_∞ coherence.

2. THE VIRASORO BAR COMPLEX

The ordered bar complex $\bar{B}^{\text{ord}}(\text{Vir}_c) = T^c(s^{-1}\overline{\text{Vir}_c})$ is an E_1 -chiral coassociative coalgebra: the bar differential $d_{\bar{B}}$ comes from the OPE residues at the boundary strata of $\text{FM}_k(\mathbb{C})$, and the deconcatenation coproduct $\Delta: \bar{B} \rightarrow \bar{B} \otimes \bar{B}$ is the cofree tensor coalgebra structure on $T^c(s^{-1}\overline{\text{Vir}_c})$. The bar complex is a single-coloured E_1 coalgebra; it is not an $\text{SC}^{\text{ch}, \text{top}}$ -coalgebra (the Swiss-cheese structure emerges on the derived centre pair $(C_{\text{ch}}^\bullet(\text{Vir}_c, \text{Vir}_c), \text{Vir}_c)$, not on $\bar{B}$).

2.1. The A_∞ operations. The λ -bracket (??) determines the binary A_∞ operation m_2 directly. The quartic pole forces m_2 to be non-associative; the Stasheff identity then forces all higher operations $m_3, m_4, \dots$ in perpetuity: the tower is infinite, and Vir_c is class **M**.

Definition 2.1 (The binary operation). The binary A_∞ operation for Vir_c is the λ -bracket itself:

$$m_2(T, T; \lambda) = \{T_\lambda T\} = \frac{c}{12} \lambda^3 + 2T \lambda + \partial T.$$

Proposition 2.2 (The Virasoro associator). The associator $A_3 = m_2 \circ (\text{id} \otimes m_2) - m_2 \circ (m_2 \otimes \text{id})$ is

$$A_3(T, T, T; \lambda_{12}, \lambda_{23}) = -\partial^2 T - (2\lambda_{12} + 3\lambda_{23}) \partial T - 2\lambda_{23}(2\lambda_{12} + \lambda_{23}) T - \frac{c}{12} \lambda_{23}^3 (2\lambda_{12} + \lambda_{23}).$$

In particular $A_3 \neq 0$ whenever $c \neq 0$ or the field-dependent terms are non-trivial: the λ -bracket is not associative. For affine Kac–Moody algebras (where the quartic pole vanishes), $A_3 = 0$ and m_2 is strictly associative.

Proof. The PVA Jacobi identity gives $A_3(\ell, \mu) = -\{T_\mu \{T_\ell T\}\}$. Direct computation using sesquilinearity of the λ -bracket: the inner bracket $\{T_\ell T\} = \partial T + 2T\ell + (c/12)\ell^3$ is substituted into the outer bracket using $\{T_\mu(\partial T)\} = -(\mu + \partial)\{T_\mu T\}$ (the left sesquilinearity identity) and $\{T_\mu T\} = \partial T + 2T\mu + (c/12)\mu^3$. The scalar term $\{T_\mu(c/12)\ell^3\} = 0$ (the bracket of a constant vanishes). Collecting terms yields (??). $\square$

The Stasheff identity $m_3 = -A_3$ then gives the ternary operation.

Proposition 2.3 (The ternary operation). The ternary A_∞ operation for Vir_c is

$$m_3(T, T, T; \lambda_{12}, \lambda_{23}) = \partial^2 T + (2\lambda_{12} + 3\lambda_{23}) \partial T + 2\lambda_{23}(2\lambda_{12} + \lambda_{23}) T + \frac{c}{12} \lambda_{23}^3 (2\lambda_{12} + \lambda_{23}).$$

The four terms encode: the second-order geometric correction ($\partial^2 T$), the first-order response $((2\lambda_{12} + 3\lambda_{23})\partial T)$, the stress-tensor backreaction $(2\lambda_{23}(2\lambda_{12} + \lambda_{23})T)$, and the three-graviton coupling $((c/12)\lambda_{23}^3(2\lambda_{12} + \lambda_{23}))$.

Proposition 2.4 (The quaternary operation). The quaternary A_∞ operation, determined by the degree-4 Stasheff identity on $\text{FM}_4(\mathbb{C}) \cong K_4$ (five compositions: two from $(r, s) = (2, 3)$, three from $(r, s) = (3, 2)$), is

$$\begin{aligned} m_4(T^4; \lambda_{12}, \lambda_{23}, \lambda_{34}) &= (\lambda_{23} + 2\lambda_{34}) \partial^2 T \\ &+ (5\lambda_{12}^2 - 4\lambda_{12}\lambda_{23} + 3\lambda_{12}\lambda_{34} - 5\lambda_{23}^2 + 10\lambda_{23}\lambda_{34} + 4\lambda_{34}^2) \partial T \\ &+ (\text{degree-3 polynomial}) \cdot T \\ &+ \frac{c}{12} Q_4(\lambda_{12}, \lambda_{23}, \lambda_{34}), \end{aligned}$$

where Q_4 is a homogeneous polynomial of degree 5. At the symmetric point $\lambda_{12} = \lambda_{23} = \lambda_{34} = \lambda$:

$$m_4(T^4; \lambda, \lambda, \lambda) = 3\lambda \partial^2 T + 13\lambda^2 \partial T + 14\lambda^3 T + \frac{7c}{12} \lambda^5.$$

The leading term is $\partial^2 T$ (at spectral depth 2), not $\partial^3 T$ (depth 1): the depth-1 coefficient vanishes by palindromic cancellation among the Stasheff compositions.

Proof. Solve the degree-4 Stasheff relation using the sesquilinearity of m_2 (??) and m_3 (??): the first-slot rule $m_3(\partial^j T, \dots) = (-\lambda_{12})^j m_3(T, \dots)$ and the last-slot rule $m_3(\dots, \partial^j T) = (\lambda_{23} + \partial)^j m_3(\dots, T)$ reduce every composition to a polynomial in $\lambda_{12}, \lambda_{23}, \lambda_{34}$. The symmetric-point evaluation (??) is verified by direct computation. $\square$

2.2. The collision kernel and its represented r -matrix. The collision kernel is obtained from the λ -bracket by the binary residue construction. A represented classical r -matrix arises under $H_{\text{CYBE}}^{\text{rep}}$.

Proposition 2.5 (Virasoro collision residue). *The collision residue of the Virasoro λ -bracket is*

$$K_{\text{Vir}_c}^{\text{coll}}(z) = \frac{c/2}{z^3} + \frac{2T}{z},$$

a cubic-plus-simple rational function. The OPE has a quartic pole; the $d \log$ absorption reduces the order by one. The Laplace kernel (without $d \log$ absorption) is $r^L(z) = (c/2)/z^4 + 2T/z^2 + \partial T/z$.

Remark 2.6 (Convention). Two conventions coexist in the literature. The collision residue (??) absorbs $d \log(z)$ (one pole fewer than OPE). The Laplace kernel $r^L(z)$ retains the full OPE poles. We use the collision residue throughout, following the bar-complex convention where the propagator is $d \log E(z, w)$ (weight 1 always). At $c = 0$ the scalar part of r^{coll} vanishes: the abelian limit, consistent with $\kappa(\text{Vir}_0) = 0$.

Proposition 2.7 (Represented Yang–Baxter equation). *Assume a representation ρ and $H_{\text{CYBE}}^{\text{rep}}(\text{Vir}_c, \rho)$. Then The CYBE*

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] = 0$$

holds for $r = \rho(K_{\text{Vir}_c}^{\text{coll}})$ from (??). It is the projection to the binary collision stratum of the Arnold relation $\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{13} + \omega_{13} \wedge \omega_{12} = 0$ on $\text{FM}_3(\mathbb{C})$, with $\omega_{ij} = d \log(z_i - z_j)$.

2.3. The c -linearity theorem. A structural pattern persists across all computed degrees m_2 through m_5 : the central charge enters each operation through a single scalar contact term, while all stress-tensor-dependent coefficients are c -independent. This separates the universal gravitational backreaction from the coupling-dependent scattering amplitude.

Theorem 2.8 (c -linearity). *For all $k \geq 2$, the scalar part of $m_k(T^{\otimes k}; \lambda_1, \dots, \lambda_{k-1})$ is **linear** in c :*

$$m_k|_{\text{scalar}} = \frac{c}{12} P_k(\lambda_1, \dots, \lambda_{k-1}),$$

where P_k is a homogeneous polynomial of degree $k + 1$. All T -dependent coefficients (the polynomials multiplying $\partial^j T$ for $0 \leq j \leq k - 2$) are independent of c .

Proof. The central charge enters the λ -bracket (??) as a scalar contact term $(c/12)\lambda^3$. Scalars are annihilated by the bracket in both slots: $\{1_\lambda T\} = 0 = \{T 1_\lambda\}$ (PVA vacuum axiom). Induction on the Stasheff relations: each m_k is a sum of compositions of lower m_j , and the only c -dependent input is the scalar part of m_2 . Since scalars are killed by subsequent brackets, c propagates linearly through the recursion. The degree count follows from $\deg P_2 = 3$ and the Stasheff composition raising degree by 2 at each step. $\square$

Remark 2.9 (Gravitational interpretation). The theorem states that graviton self-interaction at every degree is controlled by a single coupling: $c/12 = \kappa/6$, where $\kappa(\text{Vir}_c) = c/2$. The stress-tensor backreaction ($T, \partial T, \dots, \partial^{k-2} T$ terms) is universal, independent of the cosmological constant. The pure-graviton scattering amplitude is proportional to $c/12$: Newton’s constant enters once.

2.4. Infinite tower and class M.

Theorem 2.10 (Non-truncation). *The A_∞ tower of Vir_c does not truncate: $m_k \neq 0$ for all $k \geq 3$ and generic c . More precisely:*

- (i) *The quartic OPE pole forces $A_3 \neq 0$, hence $m_3 = -A_3 \neq 0$.*
- (ii) *The Stasheff identity at each subsequent degree forces $m_k \neq 0$ from the non-vanishing of the wheel integral on the associahedron K_k .*
- (iii) *At $c = 0$, the scalar terms vanish but the field-dependent terms ($\partial^j T$ coefficients) survive unchanged: the tower remains infinite even without central charge.*

The discriminant $\Delta = 8\kappa S_4 = 8(c/2) \cdot S_4 \neq 0$ for generic c , confirming class **M**: infinite shadow depth.

Proof. Part (i): the associator (??) contains the field-independent term $-\partial^2 T$, which is nonzero regardless of c , and the c -dependent term $-(c/12)\lambda_{23}^3(2\lambda_{12} + \lambda_{23})$, which is nonzero for $c \neq 0$.

Part (ii): the degree- k Stasheff identity is $\sum_{i+j=k+1} (-1)^{i(j-1)} m_i \circ m_j = 0$. The compositions involve the previously established nonzero operations $m_2, \dots, m_{k-1}$, and the obstruction at degree k is nonzero by the wheel integral computation: the Hamiltonian cycle on the complete graph K_k with $k-1$ cubic vertices has nonzero residue from the quartic OPE pole.

Part (iii): by the c -linearity theorem (Theorem ??), setting $c = 0$ kills only the scalar contact terms. The field-dependent coefficients are c -independent and nonzero (the self-coupling $T_{(1)} T = 2T$ drives the induction). $\square$

Remark 2.11 (The Drinfeld–Sokolov mechanism). The infinite A_∞ tower is manufactured by Drinfeld–Sokolov reduction from the *finite* Kac–Moody tower. The principal DS reduction of $V_k(\mathfrak{sl}_2)$ produces Vir_c with $c = 1 - 6(k+1)^2/(k+2)$, transporting the complexity class from **L** (affine: $m_k = 0$ for $k \geq 3$) to **M** (Virasoro: all $m_k \neq 0$). The mechanism: the DS stress tensor $T^{\text{DS}} = J^f + (J^h)^2/(4(k+2)) + \frac{1}{2}\partial J^h$ contains a quadratic Sugawara term whose double Wick self-contraction generates the quartic OPE pole, absent in the original affine OPE. The manufactured quartic pole makes m_2 non-associative, which seeds the entire infinite tower.

2.5. The deconcatenation coproduct. The bar complex $\bar{B}^{\text{ord}}(\text{Vir}_c) = T^c(s^{-1}\overline{\text{Vir}_c})$ carries the deconcatenation coproduct

$$\Delta[s^{-1}a_1 | \dots | s^{-1}a_n] = \sum_{i=0}^n [s^{-1}a_1 | \dots | s^{-1}a_i] \otimes [s^{-1}a_{i+1} | \dots | s^{-1}a_n],$$

which is the cofree coassociative coalgebra structure. Together with the bar differential $d_{\bar{B}}$, this makes $\bar{B}^{\text{ord}}(\text{Vir}_c)$ an E_1 dg coassociative coalgebra. The coproduct is structural (it exists for any bar complex); the gravitational content is in the differential $d_{\bar{B}}$, which encodes the A_∞ operations via the OPE residues.

The Drinfeld–Sokolov transfer theorem shows that the *spectral* coproduct Δ_z (the deformation of the deconcatenation by the spectral parameter) is primitive on Vir_c : $\Delta_z(T) = \tau_z(T) \otimes 1 + 1 \otimes T$, with no corrections at any order. The mechanism is a ghost-number obstruction: the homotopy h of the DS strong deformation retract has ghost degree -1 , and every tree in the HPL expansion of the transferred coproduct at degree $n \geq 2$ contains an uncompensated h -insertion that deposits the intermediate state in a ghost sector invisible to the projection. Gravitons do not split.

3. THE DERIVED CHIRAL CENTRE AND THE GRAVITATIONAL COMPARISON

The bar complex computes the boundary data. The level-3 closed-sector object is the derived chiral centre $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) = C_{\text{ch}}^\bullet(\text{Vir}_c, \text{Vir}_c)$: the chiral Hochschild cochain complex of the boundary algebra, defined via the chiral endomorphism operad $\text{End}_{\text{Vir}_c}^{\text{ch}}$ with spectral parameters from $\text{FM}_k(\mathbb{C})$. A gravitational bulk is obtained from it only after the Virasoro open–closed comparison map has been supplied.

3.1. Chiral Hochschild cohomology of the Virasoro algebra.

Theorem 3.1 (Bounded Virasoro cohomology and chart transport). *Type signature. Quadrant: Open. Presentation: bounded vertex cochains followed by a weight-completed curve chart. Level: 3 (derived chiral centre as universal closed sector). Hypothesis package: the Bakalov–De Sole–Kac bounded complex and a bounded-to-chart quasi-isomorphism*

$$\chi_{\text{Vir}_c}^{\text{bd}} : C_{\text{ch},b}^\bullet(\text{Vir}_c, \text{Vir}_c) \xrightarrow{\sim} C_{\text{ch}}^\bullet(\mathcal{A}_{\text{Vir}_c}, \mathcal{A}_{\text{Vir}_c}).$$

Bakalov–De Sole–Kac compute, for every central charge c ,

$$\dim H_{\text{ch},b}^n(\text{Vir}_c, \text{Vir}_c) = \begin{cases} 1, & n \in \{0, 2, 3\}, \\ 0, & n \in \mathbb{Z}_{\geq 0} \setminus \{0, 2, 3\}. \end{cases}$$

Hence $P_{\text{Vir}_c}^{\text{bd}}(t) = 1 + t^2 + t^3$. The map $\chi_{\text{Vir}_c}^{\text{bd}}$ transports this dimension vector to the curve-chart chiral Hochschild complex.

Proof. The bounded calculation is [BDSK21, Theorem 7.2]. Cohomology is invariant under the displayed quasi-isomorphism, which proves the chart statement. $\square$

Corollary 3.2 (The Virasoro universal closed sector). *Under $\chi_{\text{Vir}_c}^{\text{bd}}$, the cohomology of the universal closed-sector dg algebra*

$$Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) = C_{\text{ch}}^{\bullet}(\text{Vir}_c, \text{Vir}_c)$$

has Hilbert series $1 + t^2 + t^3$. A Virasoro open–closed quasi-isomorphism $\Psi_{\text{OCA}}: \mathcal{O}_{\text{bulk}}^{\text{phys}} \xrightarrow{\sim} Z_{\text{ch}}^{\text{der}}(\text{Vir}_c)$ transports this dg algebra to physical bulk observables. The classical absence of local propagating gravitons follows separately from the Chern–Simons presentation of three-dimensional gravity.

Proof. Theorem ?? gives the cohomology statement. The second assertion is transport along Ψ_{OCA} . Achúcarro–Townsend and Witten identify the classical gravitational field equations with flatness of the Chern–Simons connection [AT86, Wit88]. $\square$

3.2. The Swiss-cheese structure and E_3 -topologisation. The chiral derived centre carries a natural $\text{SC}^{\text{ch},\text{top}}$ structure: the pair $(Z_{\text{ch}}^{\text{der}}(\text{Vir}_c), \text{Vir}_c)$ is a two-coloured datum where the bulk (closed colour) acts on the boundary (open colour). This is the algebraic encoding of the bulk-to-boundary map in $\text{AdS}_3/\text{CFT}_2$.

Theorem 3.3 (E_3 -topologisation for Virasoro). *Type signature. Quadrant: Open. Presentation: cohomological + weight-completed (chain-level on the original direct-sum complex is Frontier F_3 , conditional on the Felder BRST Mittag–Leffler tower). Level: 3 (derived chiral centre). Hypothesis package: non-critical level $k \neq -2$ on the affine side; Sugawara primitive G_{Sug} with $[Q_{\text{tot}}, G_{\text{Sug}}] = T_{\text{Sug}}$ on cohomology; A_{∞} coherence of the topologisation primitives in the chain-level lift (open at Class **M**).*

For Vir_c obtained by Drinfeld–Sokolov reduction from $V_k(\mathfrak{sl}_2)$ at non-critical level $k \neq -2$ (the critical level $k = -h^{\vee} = -2$ maps to $c \rightarrow \infty$ under $c = 1 - 6(k+1)^2/(k+2)$; the condition excludes this singularity), the $\text{SC}^{\text{ch},\text{top}}$ structure on the derived centre pair promotes to an E_3 -topological structure on $H^{\bullet}(Z_{\text{ch}}^{\text{der}}(\text{Vir}_c), Q)$. The mechanism: the Sugawara stress tensor satisfies $T(z) = \{Q_{\text{CS}}, G'(z)\}$ in the BRST complex, so $\mathbb{C}$ -translations are Q -exact, the two $\text{SC}^{\text{ch},\text{top}}$ colours collapse in cohomology, and $E_2^{\text{hol}} \times E_1^{\text{top}} \rightarrow E_2^{\text{top}} \times E_1^{\text{top}} = E_3^{\text{top}}$ via Dunn additivity.

Remark 3.4 (Scope). The E_3 -topological structure is proved cohomologically: it holds on $H^{\bullet}(Z^{\text{der}}, Q)$, not necessarily at the cochain level. For class **M** algebras (Virasoro), the chain-level E_3 is conditional on A_{∞} coherence of the topologisation. The cohomological statement suffices for the genus-tower computations of Section ??. The proof is for affine Kac–Moody $V_k(\mathfrak{sl}_2)$ at non-critical level, transported to Virasoro by the DS intertwining theorem; general chiral algebras with conformal vector are conjectural.

3.3. The bulk-to-boundary map. The bulk-to-boundary map sends the degree-2 chiral Hochschild class $\Theta_c \in \text{ChirHoch}^2$ to the stress tensor $T \in \text{Vir}_c$ via the conformal Ward identity:

$$\beta_B(\partial_c)(z_o) \cdot T(w) = \frac{T(w)}{(z_o - w)^2} + \frac{\partial T(w)}{z_o - w}.$$

These constructions form the algebraic closed-sector/boundary/line comparison diagram. The bar complex supports three typed functors:

- (i) $\Omega(\bar{B}) \simeq \text{Vir}_c$ (cobar recovers the boundary; bar-cobar adjunction),
- (ii) $K_X(\text{Vir}_c) := \mathbb{D}_{\text{Ran}} \bar{B}_X^{\text{ch}}(\text{Vir}_c)$ is the full-bar Verdier algebra, with a chosen comparison $\nu_c: K_X(\text{Vir}_c) \rightarrow \text{Vir}_c^!$;
- (iii) $\text{RHom}_{\text{Vir}_c^e}(\text{Vir}_c, \text{Vir}_c) = C_{\text{ch}}^\bullet(\text{Vir}_c, \text{Vir}_c)$ is the universal algebraic closed sector. An OCA realization Ψ_{OCA} supplies its physical bulk interpretation.

4. THE SHADOW TOWER AS ALGEBRAIC-SECTOR TRACE

The universal Maurer–Cartan element Θ_{Vir_c} in the modular convolution algebra $\mathfrak{g}_{\text{Vir}_c}^{\text{mod}}$ satisfies the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ and encodes the algebraic-sector trace of the perturbative expansion of 3d gravity (operator $\rightarrow$ scalar trace direction; level-4 $\rightarrow$ level-5 of the Beilinson tower). Its projections to successive degrees constitute the *shadow obstruction tower*, a sequence of invariants $\{S_r\}_{r \geq 2}$ that record the perturbative corrections to the semiclassical gravitational dictionary at the level of the algebraic sector. The inverse construction (from the $\{S_r\}$ to a compact dynamical-metric path integral) is the named open obstruction Fiz ; here we compute the trace.

4.1. The shadow invariants.

Definition 4.1 (Shadow tower of Vir_c). The shadow obstruction tower is the sequence $S_2, S_3, S_4, \dots$ of projections of the MC element Θ_{Vir_c} to successive degrees in the convolution algebra. The first four named invariants are:

- (i) $S_2 = \kappa(\text{Vir}_c) = c/2$: the curvature of the bar complex (UNIFORM-WEIGHT).
- (ii) $S_3 = \mathfrak{C}(\text{Vir}_c) = -c$: the cubic shadow, encoding the Lie bracket structure of the convolution algebra.
- (iii) S_4 : the quartic shadow, which determines the quartic contact invariant

$$\mathfrak{Q}_{\text{Vir}}^{\text{contact}} = \frac{\text{IO}}{c(5c + 22)}.$$

The numerator IO arises from the Shapovalov inner product at level 4; the denominator $c(5c + 22)$ is the Kac determinant. Poles at $c = 0$ ($G = \infty$) and $c = -22/5$ (Lee–Yang) mark reorganisation points of the tower.

- (iv) S_r for $r \geq 5$: the higher shadow obstructions, each determined by the degree- r Stasheff identity and the Kac determinant at levels $\leq r$.

4.2. The discriminant and class **M**.

Theorem 4.2 (Discriminant). *The discriminant of Vir_c is*

$$\Delta(\text{Vir}_c) = 8\kappa S_4 = 8 \cdot \frac{c}{2} \cdot S_4 \neq 0$$

for generic c (in particular for all $c > 0$). The discriminant is linear in κ , not quadratic. Since $\Delta \neq 0$, the shadow tower does not terminate: Vir_c is class **M** with infinite shadow depth $r_{\text{max}} = \infty$.

Proof. The quartic OPE pole forces $S_4 \neq 0$ (Proposition ??): the scalar contact term $(7c/12)\lambda^5$ at the symmetric point is proportional to c and nonzero for $c \neq 0$. Combined with $\kappa = c/2 \neq 0$ for $c \neq 0$: $\Delta = 8(c/2)S_4 \neq 0$. When $\Delta \neq 0$, the recursion governing the shadow tower propagates to all genera, giving $r_{\text{max}} = \infty$. $\square$

4.3. **Gravitational interpretation of the shadow degrees.** Each shadow degree has a direct gravitational reading. The table:

Shadow	Algebraic content	Gravitational counterpart
$\kappa = c/2$	bar complex curvature	Brown–Henneaux central charge; one-loop Weyl anomaly
$\mathfrak{C} = -c$	cubic Lie bracket	cubic graviton vertex $\propto \ell/(16\pi G)$
$\mathfrak{Q} = \frac{10}{c(5c+22)}$	quartic contact	four-graviton amplitude; genus-2 contact correction
$S_r, r \geq 5$	degree- r obstruction	r -graviton vertex; higher-loop correction

4.4. **The genus tower.** The curvature $\kappa = \kappa(\text{Vir}_c) = c/2$ controls the scalar genus shadow of the gravitational partition function through the fiberwise differential $d_{\text{fib}}^2 = \kappa \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_g$.

Theorem 4.3 (The intrinsic genus tower). *The intrinsic genus- g free energy of Vir_c is (UNIFORM-WEIGHT, all $g \geq 1$):*

$$F_g^{\text{intr}}(\text{Vir}_c) = \kappa(\text{Vir}_c) \cdot \lambda_g^{\text{FP}}, \quad \lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_g} \lambda_g = \frac{2^{2g-1} - 1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!} > 0.$$

The generating function is the $\hat{A}$ -genus:

$$\sum_{g \geq 1} F_g^{\text{intr}} x^{2g} = \frac{c}{2} (\hat{A}(ix) - 1), \quad \hat{A}(x) = \frac{x/2}{\sinh(x/2)}.$$

The first two terms:

$$F_1^{\text{intr}} = \frac{\kappa}{24} = \frac{c}{48},$$

$$F_2^{\text{intr}} = \frac{7\kappa}{5760} = \frac{7c}{11520}.$$

Proof. The genus- g free energy is $F_g^{\text{intr}} = \kappa \cdot \int_{\overline{\mathcal{M}}_g} \lambda_g$ by the scalar genus tower theorem applied to the single-generator algebra Vir_c (uniform weight: one generating field of conformal weight 2). The Hodge integrals λ_g^{FP} are the Faber–Pandharipande values, whose generating function is $\sum_{g \geq 1} \lambda_g^{\text{FP}} x^{2g} = \hat{A}(ix) - 1 = (x/2)/\sinh(x/2) - 1$ by the Mittag-Leffler expansion $(x/2)/\sinh(x/2) = 1 + 2 \sum_{g \geq 1} \frac{(2^{2g-1} - 1)|B_{2g}|}{(2g)!} x^{2g}$. The substitution $x \mapsto ix$ converts $\sinh(ix/2) = i \sin(x/2)$, producing the $\hat{A}(ix)$ form with all positive coefficients (the Wick rotation from Lorentzian to Euclidean signature).

For F_1^{intr} : $\lambda_1^{\text{FP}} = (2^1 - 1)|B_2|/(2^1 \cdot 2!) = 1 \cdot (1/6)/(2 \cdot 2) = 1/24$, using $B_2 = 1/6$; then $F_1^{\text{intr}} = (c/2)(1/24) = c/48$.

For F_2^{intr} : $\lambda_2^{\text{FP}} = (2^3 - 1)|B_4|/(2^3 \cdot 4!) = 7/(8 \cdot 24) \cdot (1/30)$; since $B_4 = -1/30$, $|B_4| = 1/30$, $\lambda_2^{\text{FP}} = 7 \cdot (1/30)/(8 \cdot 24) = 7/(5760)$. Then $F_2^{\text{intr}} = (c/2)(7/5760) = 7c/11520$. $\square$

Remark 4.4 (Why the $\hat{A}$ -genus, not Todd). Three independent reasons force the $\hat{A}$ -genus: (a) *symmetry*: the genus tower uses only even powers of x ; the Todd class has odd powers; (b) *KO orientation*: the Hodge bundle $\mathbb{E}$ on $\overline{\mathcal{M}}_g$ carries a real structure via Serre duality ($\mathbb{E}^* \simeq \mathbb{E} \otimes \omega$); real vector bundles are oriented by KO-theory, whose multiplicative genus is $\hat{A}$; (c) *explicit integration*: the Faber–Pandharipande integrals match the $\hat{A}(ix)$ coefficients through $g = 6$.

Remark 4.5 (Intrinsic versus effective). The intrinsic free energies $F_g^{\text{intr}} = \kappa \cdot \lambda_g^{\text{FP}}$ use $\kappa(\text{Vir}_c) = c/2$ and are the fundamental algebraic invariants. The *effective* free energies incorporate ghost subtraction from the bc ghost system ($c_{\text{ghost}} = -26$, $\kappa_{\text{ghost}} = -13$):

$$F_g^{\text{eff}} = \kappa_{\text{eff}} \cdot \lambda_g^{\text{FP}}, \quad \kappa_{\text{eff}} = \kappa + \kappa_{\text{ghost}} = \frac{c - 26}{2}.$$

The cancellation $\kappa_{\text{eff}} = 0$ at $c = 26$ is matter-ghost cancellation (the bosonic string critical point), not vanishing of the intrinsic curvature ($\kappa(\text{Vir}_{26}) = 13 \neq 0$). Throughout this paper, F_g without superscript denotes F_g^{intr} ; the effective version appears only in contexts involving ghost coupling.

4.5. The quartic contact correction at genus 2.

Proposition 4.6 (Genus-2 contact correction). *The quartic contact shadow (??) contributes a genus-2 correction via the $r = 4$ planted-forest graph amplitude:*

$$F_2^{\text{contact}} = \mathfrak{Q} \cdot \int_{\overline{M}_2} \psi_1^2 = \frac{10}{c(5c + 22)} \cdot \frac{1}{240} = \frac{1}{24 c(5c + 22)},$$

where $\int_{\overline{M}_2} \psi_1^2 = 1/240$ by Faber's formula. In the semiclassical limit $c \rightarrow \infty$: $F_2^{\text{contact}} \sim 1/(120 c^2)$.

Remark 4.7 (Convergence dichotomy). The shadow tower exhibits a sharp analytic dichotomy: the *scalar sector* (contact terms $\propto c/12$) has an algebraic generating function $H(t) = t^2 \sqrt{Q_{\text{Vir}}(t)}$ with the shadow metric $Q_{\text{Vir}}(t) = c^2 + 12ct + \frac{180c+872}{5c+22} t^2$; its Taylor series converges for $|t| < R_{\text{scal}} = c\sqrt{(5c+22)/(180c+872)}$. The *field sector* (T -dependent terms, c -independent) diverges factorially: $\|m_k|_T\| \sim C^k \cdot k!$ (Gevrey-1), requiring Borel resummation. The Borel sum is unambiguous because the spectral curve $u^2 = Q_{\text{Vir}}(t)$ has genus 0 (no real branch points for $c > 0$): 3d gravity has no resurgent ambiguity.

5. KOSZUL DUALITY AS THE ALGEBRAIC SHADOW OF S-DUALITY

The Virasoro algebra has a Koszul dual: it is another Virasoro algebra, at the complementary central charge $26 - c$. This algebraic duality is the level-2 / level-5 algebraic shadow of gravitational S-duality, computed by trace + clutching on the algebraic sector; the self-dual point $c = 13$ is the fixed point of the Koszul involution. Lifting this algebraic shadow to a nonperturbative S-duality on the dynamical-metric path integral is part of F12 (Section ??).

5.1. The Koszul dual.

Conjecture 5.1 (Virasoro reflection comparison). *Fix a completed Virasoro presentation and its Verdier/Koszul comparison $\nu_c: K_X(\text{Vir}_c) \rightarrow \text{Vir}_c^!$. The proposed reflection comparison is a quasi-isomorphism*

$$\chi_c: \text{Vir}_c^! \xrightarrow{\sim} \text{Vir}_{26-c}.$$

Its scalar consequence is

$$K(\text{Vir}) = \kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = \frac{c}{2} + \frac{26-c}{2} = 13.$$

Self-duality holds at $c = 13$: $\text{Vir}_{13}^! \simeq \text{Vir}_{13}$ after choosing the coherence of χ_{13} .

Remark 5.2 (Comparison obligation). The scalar identity in (??) follows directly from the two displayed central-charge parameters. The categorical statement (??) requires a chain map on the completed Virasoro bar model, compatibility with the Virasoro relations, and a quasi-isomorphism proof. These three constructions constitute the reflection-comparison obligation.

5.2. **The two critical central charges.** The Koszul involution $c \mapsto 26 - c$ has two distinguished points.

Proposition 5.3 (Two critical points).

- (i) $c = 13$: **Koszul self-duality.** $\text{Vir}_{13}^! = \text{Vir}_{13}$. The boundary algebra equals its Koszul dual: $\kappa = \kappa' = 13/2$. The genus tower is symmetric under the involution: $F_g(c) = F_g(26 - c)$ for all $g \geq 1$. The graviton resolvent $\mathcal{G}(t; \lambda)$ acquires a $c \mapsto 26 - c$ symmetry. This is the algebraic self-duality of 3d gravity; it is not the string theory critical point.
- (ii) $c = 26$: **matter-ghost cancellation.** The effective curvature $\kappa_{\text{eff}} = (c - 26)/2 = 0$: the effective genus tower $F_g^{\text{eff}} = 0$ for all $g \geq 1$ (the intrinsic tower $F_g^{\text{intr}} = 13 \lambda_g^{\text{FP}} \neq 0$). The gravitational partition function becomes S -invariant (modular weight vanishes). This is the bosonic string critical point. The Koszul dual $\text{Vir}_{26}^! = \text{Vir}_0$ is the Witt algebra at $c = 0$: uncurved ($\kappa = 0$) but non-formal (infinite A_∞ tower from the field-dependent terms).

Proof. Part (i): at $c = 13$, the involution $c \mapsto 26 - c$ fixes c , so $\text{Vir}_{13}^! = \text{Vir}_{26-13} = \text{Vir}_{13}$. The symmetry $F_g(c) = F_g(26 - c)$ follows from $F_g = (c/2)\lambda_g^{\text{FP}}$: replacing c by $26 - c$ gives $((26 - c)/2)\lambda_g^{\text{FP}}$; the two are equal when $c/2 = (26 - c)/2$, i.e. $c = 13$.

Part (ii): at $c = 26$, $\kappa_{\text{eff}} = (c - 26)/2 = 0$ directly, so $F_g^{\text{eff}} = 0$. The effective genus-1 partition function $Z_1^{\text{eff}}(\tau) = \eta(\tau)^{-\kappa_{\text{eff}}} = \eta(\tau)^0 = 1$. The Koszul dual is Vir_0 ; by Theorem ??(iii), Vir_0 has $\kappa = 0$ (uncurved) but $m_k \neq 0$ for all $k \geq 3$ (non-formal). $\square$

5.3. **The complementarity constant and Page time.** The complementarity constant $K = 13$ has physical content beyond its algebraic origin.

Proposition 5.4 (Complementarity and the genus tower). For any $g \geq 1$:

$$F_g(\text{Vir}_c) + F_g(\text{Vir}_{26-c}) = 13 \cdot \lambda_g^{\text{FP}}.$$

The sum is independent of c : the total intrinsic genus- g free energy of the Koszul pair is a topological constant equal to the complementarity constant $K = 13$ times the Hodge integral.

Proof. $F_g(\text{Vir}_c) + F_g(\text{Vir}_{26-c}) = \frac{c}{2}\lambda_g^{\text{FP}} + \frac{26-c}{2}\lambda_g^{\text{FP}} = 13 \lambda_g^{\text{FP}}$. $\square$

Remark 5.5 (Verdier duality and the modular S -transform). The Koszul involution $\mathbb{D}: c \mapsto 26 - c$ (Verdier duality on the bar complex) and the modular S -transform $S: \tau \mapsto -1/\tau$ commute on genus-1 MC elements: $S \circ \mathbb{D}(\Theta^{(1)}(\tau)) = \mathbb{D} \circ S(\Theta^{(1)}(\tau)) = \Theta_{\text{Vir}_{26-c}}^{(1)}(-1/\tau)$. They act on different variables (c vs. τ). The commutativity produces a paired Cardy formula: $S_{\text{Cardy}}(\text{Vir}_c; h) + S_{\text{Cardy}}(\text{Vir}_{26-c}; h) = 2\pi\sqrt{ch/6} + 2\pi\sqrt{(26-c)h/6}$, which is maximised at $c = 13$ by concavity of $\sqrt{\cdot}$.

5.4. **The quartic contact under Koszul duality.**

Proposition 5.6 (Quartic complementarity). Under the Koszul involution $c \mapsto 26 - c$:

$$\mathfrak{Q}(c) + \mathfrak{Q}(26 - c) = \frac{10}{c(5c + 22)} + \frac{10}{(26 - c)(152 - 5c)}.$$

At $c = 13$ (self-dual): $\mathfrak{Q}(13) = 10/(13 \cdot 87) = 10/1131$. The quartic contact invariant is the first nonlinear witness of the duality between boundary and line sectors.

Remark 5.7 (The gauge-gravity-matter trichotomy). The complexity classification by the discriminant Δ organises the standard landscape into three columns:

	Gauge (CS/KM)	Pure gravity (Vir/ W_N)	Matter-coupled (BP)
Class	L	M	M
Shadow depth	3	∞	∞
$\mathcal{A}_\infty$ tower	finite	infinite	infinite
Δ	0	$\neq 0$	$\neq 0$

The passage from gauge ($\Delta = 0$, integrable) to gravity ($\Delta \neq 0$, chaotic) is Drinfeld–Sokolov reduction: the Sugawara construction manufactures the quartic pole from double poles, and **L** $\rightarrow$ **M**.

6. THE HOLOGRAPHIC DATUM $\mathcal{H}(\text{Vir}_c)$

The six preceding sections assembled the gravitational Koszul data from the single input (??). We now organise the output into the *holographic modular Koszul datum*: a seven-entry algebraic package. It becomes a physical bulk datum only after the open–closed comparison and the line-comparison maps are supplied.

6.1. The seven-entry package.

Definition 6.1 (Holographic datum of Vir_c). The *holographic datum* of 3d gravity at central charge $c = 3\ell/(2G)$ is the seven-entry package

$$\mathcal{H}(\text{Vir}_c) = (\text{Vir}_c, \text{Vir}_c^i, \text{Vir}_c^!, Z_{\text{ch}}^{\text{der}}(\text{Vir}_c), r_c(z), \Theta_c, \nabla^{\text{hol}}),$$

where the seven components are:

- (i) Vir_c : the **boundary algebra**, the Virasoro vertex algebra at central charge $c = 3\ell/(2G)$, with OPE (??) and $\kappa(\text{Vir}_c) = c/2$;
- (ii) Fix a quadratic-linear presentation $\text{Vir}_c = T_X(V_c)/(R_c)$. Its **presentation coalgebra** is $\text{Vir}_c^i = C_X(s^{-1}V_c, s^{-2}R_c)$, with $q_c: \text{Vir}_c^i \rightarrow \tilde{B}_X^{\text{ch}}(\text{Vir}_c)$.
- (iii) The **strict Koszul partner** $\text{Vir}_c^!$ is reached from $K_X(\text{Vir}_c) = \mathbb{D}\tilde{B}_X^{\text{ch}}(\text{Vir}_c)$ through the chosen pair map $\nu_c: K_X(\text{Vir}_c) \rightarrow \text{Vir}_c^!$. Conjecture ?? formulates the further reflection comparison with Vir_{26-c} .
- (iv) $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c)$: the **algebraic closed-sector object**. The open–closed map Ψ_{OCA} realizes it in the physical bulk when the OCA comparison package holds. The line category is $\text{Vir}_c^!$ -mod on the strict line-comparison surface.
- (v) $r_c(z) = (c/2)/z^3 + 2T/z$: the **classical r -matrix** (Proposition ??), the collision residue of the Virasoro λ -bracket with cubic-plus-simple poles;
- (vi) Θ_c : the **universal Maurer–Cartan element** in the modular convolution algebra $\mathfrak{g}_{\text{Vir}_c}^{\text{mod}}$, whose projections constitute the shadow obstruction tower $\{S_r\}_{r \geq 2}$ (Definition ??);
- (vii) ∇^{hol} : the **shadow connection**, the flat connection on the configuration-space bundle determined by the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

Remark 6.2 (The algebraic closed/open/line triangle). Entries (i), (iii), and (iv) form the *algebraic closed/open/line triangle*:

$$\begin{aligned} \text{Closed sector} &= Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) = C_{\text{ch}}^\bullet(\text{Vir}_c, \text{Vir}_c) \\ \text{Boundary} &= \text{Vir}_c \\ \text{Lines} &= \text{Vir}_c^! \text{-mod (on Koszul locus)} \end{aligned}$$

The closed-to-boundary edge is the Swiss-cheese restriction $\rho_\partial: Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) \rightarrow \text{Vir}_c$. The boundary-to-lines edge is the composite of q_c , Ran-space Verdier duality, and ν_c . The OCA comparison Ψ_{OCA} maps the algebraic closed sector to the physical bulk observables.

6.2. Explicit computation of the seven-entry package. The seven-entry package $\mathcal{H}(\text{Vir}_c)$ is determined by c . Its nonformal numerical projections admit closed-form expressions.

Theorem 6.3 (The holographic datum: explicit form). *The seven entries of $\mathcal{H}(\text{Vir}_c)$ are determined by the single datum c ; their numerical projections include:*

- (i) *Boundary curvature: $\kappa(\text{Vir}_c) = c/2$; dual curvature: $\kappa(\text{Vir}_{26-c}) = (26-c)/2$; complementarity: $\kappa + \kappa' = 13$.*
- (ii) *Shadow obstruction tower through degree 5:*

$$\begin{aligned} S_2 &= \kappa = c/2 && (\text{curvature}), \\ S_3 &= -c && (\text{cubic shadow}), \\ S_4 &= \frac{10}{c(5c+22)} && (\text{quartic contact}), \\ S_5 &= \frac{-48}{c^2(5c+22)} && (\text{quintic}). \end{aligned}$$

At large c (semiclassical): $S_4 \sim 2/c^2 \sim G^2$, $S_5 \sim -48/(5c^3) \sim G^3$.

- (iii) *Classical r -matrix: $r_c(z) = (c/2)/z^3 + 2T/z$; at $c = 0$: $r_0(z) = 2T/z$ (field-dependent only, no central extension); at $c = 13$: $r_{13}(z) = (13/2)/z^3 + 2T/z$ (self-dual).*
- (iv) *Spectral coproduct: strictly primitive, $\Delta_z(T) = \tau_z(T) \otimes 1 + 1 \otimes T$. No higher terms at any degree: gravitons do not split.*
- (v) *Shadow connection: $\nabla^{\text{hol}} = d - \sum_{r \geq 2} S_r \omega_{(r)}$, where $\omega_{(r)}$ is the degree- r component of the bar-complex propagator form on the configuration space $\text{FM}_n(\mathbb{C})$. Flatness of ∇^{hol} is equivalent to the MC equation for Θ_c .*
- (vi) *Discriminant: $\Delta = 8\kappa S_4 = 40/(5c+22) \neq 0$ for $c \neq -22/5$. Class **M**: infinite shadow depth.*

Proof. Each listed projection was computed in the preceding sections: (i) equation (??); (ii) Definition ?? for S_2 – S_4 , and the quintic shadow from the shadow-metric expansion $H(t) = t^2 \sqrt{Q_{\text{Vir}}(t)}$ with $Q_{\text{Vir}}(t) = c^2 + 12ct + (180c + 872)t^2/(5c+22)$, giving $S_5 = [t^5]H/5 = -48/[c^2(5c+22)]$; (iii) equation (??); (iv) the DS ghost-number obstruction (Section ??); (v) the MC equation of Section ??; (vi) equation (??). $\square$

6.3. The shadow connection and soft graviton hierarchy. The shadow connection ∇^{hol} decomposes the MC element into a hierarchy of Ward identities. The first three are classical; the rest are new.

Proposition 6.4 (Shadow connection and soft gravitons). *The shadow connection ∇^{hol} on $\text{FM}_n(\mathbb{C})$ acts on n -point correlation functions of primary operators $\mathcal{O}_i$ of weight h_i . Its decomposition by shadow degree gives:*

- (i) *Degree 2 ($S_2 = c/2$): the Weinberg leading soft graviton theorem. Soft factor: $S_i^{(0)} = \partial_{z_i}/(z_0 - z_i)$.*
- (ii) *Degree 2 (double pole of S_2): the Cachazo–Strominger subleading soft theorem. Soft factor: $S_i^{(1)} = h_i/(z_0 - z_i)^2$.*
- (iii) *Degree 3–4: the cubic shadow $S_3 = -c$ is gauge-trivial; the first non-trivial sub-subleading contribution is the quartic contact $S_4 = 10/[c(5c+22)]$, the Cachazo–He–Yuan soft factor requiring the ternary A_∞ operation m_3 .*
- (iv) *Degree $r \geq 5$: one independent Ward identity per degree, forced by the class **M** property ($m_k \neq 0$ for all $k \geq 3$). The infinite A_∞ tower produces infinitely many soft theorems beyond the known three.*

At large c : the soft factors at degree r scale as $S_r \sim G^{r-2}$, and the semiclassical limit retains only S_2 (Weinberg) and the subleading $S^{(1)}$ (Cachazo–Strominger).

Proof. The shadow connection is flat (from the MC equation), so each degree- r component S_r gives an independent Ward identity. The degree-2 components extract the conformal Ward identity, whose simple-pole and double-pole residues are the Weinberg and Cachazo–Strominger soft factors. The cubic shadow is gauge-trivial: $S_3 = d_2 \sigma_3$ for an explicit gauge transformation σ_3 (using the fact that the Virasoro algebra is simple for generic c , so $H^1(\text{Vir}_c, \text{Vir}_c) = 0$). The quartic contact $S_4 = 10/[c(5c + 22)]$ is genuinely new by the non-vanishing of the Kac determinant at level 4. For degree ≥ 5 : non-vanishing follows from the infinite tower (Theorem ??). $\square$

6.4. The spectral braiding.

Proposition 6.5 (Virasoro fusion kernel as braiding). *The braided monoidal structure on $\mathcal{C}_c$ is determined by the r -matrix $r_c(z) = (c/2)/z^3 + 2T/z$ via the KZ-type connection on the category of Virasoro modules:*

$$\nabla_{\text{KZ}} = d - \sum_{i < j} r_c(z_i - z_j) dz_{ij},$$

whose monodromy is the Virasoro fusion kernel (the Ponsot–Teschner 6 j -symbol [PT01]). The KZ connection has an irregular singularity at $z_i = z_j$ of Poincaré rank 2 (from the cubic pole), in contrast to the regular-singular KZ connection of affine Kac–Moody algebras (where the r -matrix has only a simple pole). The irregular singularity produces Stokes phenomena in the braiding, not present in the KM setting.

Proof. The classical r -matrix (??) determines the connection 1-form $\sum r_c(z_{ij}) dz_{ij}$ on $\text{Conf}_n(\mathbb{C})$. The CYBE (Proposition ??) is equivalent to flatness of this connection. The Poincaré rank at $z_i = z_j$ is the order of the pole minus one: $(z_{ij})^{-3}$ gives rank 2. The monodromy of the flat connection around $z_j = z_i + \epsilon e^{2\pi i \theta}$ produces the braiding on modules; for the Virasoro algebra, this is the Ponsot–Teschner kernel. $\square$

Remark 6.6 (Gravity versus gauge theory). The holographic datum distinguishes gravity from gauge theory through three structural signatures:

	Gauge ($V_k(\mathfrak{g})$)	Gravity (Vir_c)
r -matrix pole	simple ($1/z$)	cubic ($1/z^3$)
Coproduct	non-trivial	primitive
$\mathcal{A}_\infty$ tower	finite ($m_k = 0, k \geq 3$)	infinite (all $m_k \neq 0$)
KZ singularity	regular	irregular (Poincaré rank 2)
Shadow depth	3 (class L)	∞ (class M)
Discriminant	$\Delta = 0$	$\Delta \neq 0$

The passage from gauge to gravity is Drinfeld–Sokolov reduction: the Sugawara construction manufactures the quartic pole from double poles, simultaneously creating the infinite $\mathcal{A}_\infty$ tower, the irregular KZ singularity, and the primitive coproduct.

7. THE GENUS EXPANSION AND THE $\hat{\mathcal{A}}$ -GENUS

The shadow obstruction tower of Section ?? produces a sequence of genus- g free energies F_g through the scalar genus tower theorem. This section derives the same free energies from an independent method: the Grothendieck–Riemann–Roch theorem applied to the Hodge bundle on $\overline{\mathcal{M}}_g$. The agreement of the two derivations is a non-trivial consistency check.

7.1. The genus-1 formal coefficient and the Weinberg soft theorem.

Theorem 7.1 (Genus-1 formal coefficient). *The genus-1 formal scalar coefficient of Vir_c is (UNIFORM-WEIGHT):*

$$F_1(\text{Vir}_c) = \frac{\kappa(\text{Vir}_c)}{24} = \frac{c}{48}.$$

Three independent derivations:

- (i) **Shadow tower.** $F_1 = \kappa \cdot \lambda_1^{\text{FP}}$, where $\lambda_1^{\text{FP}} = \int_{\overline{\mathcal{M}}_{1,1}} \lambda_1 = 1/24$ and $\kappa(\text{Vir}_c) = c/2$.
- (ii) **Dedekind eta function.** The genus-1 partition function is $Z_1(\tau) = \text{Tr}_{\text{Vir}_c}(q^{L_0 - c/24}) = q^{-c/24} / \prod_{n \geq 1} (1 - q^n) = q^{-c/24} \eta(\tau)^{-1}$, where $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ is the Dedekind eta function. The free energy $F_1 = -\log |Z_1|^2|_{q \rightarrow 0} = c/48 \cdot \log |q|^{-2} = c/48$ (leading term).
- (iii) **Soft graviton.** The Weinberg leading soft factor $S_i^{(0)} = \partial_{z_i} / (z_o - z_i)$ applied to the genus-1 one-point function gives $\langle T \rangle_{g=1} = -c/24$ (the Casimir energy on the torus). The free energy $F_1 = \kappa \cdot \lambda_1^{\text{FP}} = (c/2)(1/24) = c/48$.

Proof. Derivation (i) follows from the genus tower (Theorem ??) at $g = 1$, using the intrinsic curvature $\kappa = c/2$ (not the effective curvature).

Derivation (ii): the vacuum character of Vir_c for generic c (no null vectors) is $\chi_o(\tau) = q^{-c/24} \prod_{n \geq 2} (1 - q^n)^{-1}$. The full partition function on a genus-1 surface with modular parameter τ includes the full Fock space: $Z_1(\tau) = q^{-c/24} \prod_{n \geq 1} (1 - q^n)^{-1}$. The leading q -expansion gives the free energy $F_1 = c/48$.

Derivation (iii): the genus-1 one-point function is

$$\langle T \rangle_{g=1} = \frac{\text{Tr}(L_o q^{L_o - c/24})}{\text{Tr}(q^{L_o - c/24})} = q \partial_q \log Z_1(\tau) = -\frac{c}{24} + O(q).$$

This is the Casimir energy density. The integrated free energy is $F_1 = (c/2) \cdot (1/24)$. $\square$

7.2. The genus-2 formal coefficient.

Theorem 7.2 (Genus-2 formal coefficient). *The genus-2 formal scalar coefficient is (UNIFORM-WEIGHT):*

$$F_2(\text{Vir}_c) = \frac{7\kappa(\text{Vir}_c)}{5760} = \frac{7c}{11520}.$$

The Hodge integral value $\lambda_2^{\text{FP}} = \int_{\overline{\mathcal{M}}_2} \lambda_2 = 7/5760$ uses $B_4 = -1/30$ and $(2^3 - 1)/2^3 = 7/8$.

Proof. From the genus tower (Theorem ??): $F_2 = \kappa \cdot \lambda_2^{\text{FP}} = (c/2)(7/5760) = 7c/11520$. The computation of λ_2^{FP} : $\lambda_2^{\text{FP}} = (2^{2 \cdot 2 - 1} - 1)|B_{2,2}|/(2^{2 \cdot 2 - 1} \cdot (2 \cdot 2)!) = (2^3 - 1)|B_4|/(2^3 \cdot 4!) = 7 \cdot (1/30)/(8 \cdot 24) = 7/5760$. $\square$

Remark 7.3 (The quartic contact correction). The shadow free energy $F_2 = 7\kappa/5760$ is the scalar-lane contribution. The quartic contact shadow adds a correction (Proposition ??): $F_2^{\text{contact}} = \mathfrak{Q} \cdot \int_{\overline{\mathcal{M}}_2} \psi_1^2 = 10/[c(5c + 22)] \cdot 1/240 = 1/[24c(5c + 22)]$. At large c : $F_2^{\text{contact}} \sim 1/(120c^2)$, which is $O(G^2)$, a genuine two-loop quantum gravity correction invisible in the one-loop $\hat{A}$ -genus formula.

7.3. The GRR derivation. The $\hat{A}$ -genus formula for the genus tower admits an independent derivation from the Grothendieck–Riemann–Roch theorem, providing a second proof that does not depend on the shadow obstruction tower.

Theorem 7.4 (GRR derivation of the genus tower). *Let $\pi: C_g \rightarrow \overline{M}_g$ be the universal curve, and $\mathbb{E} = \pi_* \omega_{C_g/\overline{M}_g}$ the Hodge bundle. The Grothendieck–Riemann–Roch theorem gives:*

$$\text{ch}(\mathbb{E}) = \pi_*(\text{td}(\omega_{C_g/\overline{M}_g})) = g + \sum_{k \geq 1} \frac{B_{2k}}{(2k)!} \kappa_{2k-1},$$

where κ_{2k-1} is the Miller–Morita–Mumford class and B_{2k} are Bernoulli numbers. The generating function of the Hodge integrals $\lambda_g^{\text{FP}} = \int_{\overline{M}_g} \lambda_g$ is determined by the $\hat{A}$ -genus:

$$\sum_{g \geq 1} \lambda_g^{\text{FP}} x^{2g} = \hat{A}(ix) - 1, \quad \hat{A}(x) = \frac{x/2}{\sinh(x/2)}.$$

Proof. The Chern character of the Hodge bundle is computed by GRR: $\text{ch}(\mathbb{E}) = \pi_*(\text{td}(\Omega_\pi^1))$, where td is the Todd class. The top Chern class $\lambda_g = c_g(\mathbb{E})$ is extracted from $\text{ch}(\mathbb{E})$ via the Newton identities relating Chern characters to Chern classes. The generating function of $\lambda_g^{\text{FP}} = \int_{\overline{M}_g} \lambda_g$ is then determined by integrating the Todd class over the fibres of π : the fibre integral produces the $\hat{A}$ -genus of the tangent bundle of C_g (which has real rank 2, so the KO orientation produces $\hat{A}$ rather than Todd). $\square$

Remark 7.5 (Why $\hat{A}$, not Todd: three reasons). The GRR derivation confirms Remark ??:

- (a) *Even powers*: the genus tower uses x^{2g} (even); the Todd class has odd powers.
- (b) *KO orientation*: the Hodge bundle carries a real structure from Serre duality $\mathbb{E}^* \simeq \mathbb{E} \otimes \omega$; the real vector bundle is oriented by KO-theory, whose multiplicative genus is $\hat{A}$.
- (c) *Explicit verification*: the Faber–Pandharipande integrals match $\hat{A}(ix)$ through $g = 6$.

7.4. Uniform-weight structure for the Virasoro algebra.

Proposition 7.6 (Uniform-weight and absence of cross-channel corrections). *The Virasoro algebra Vir_c has a single generating field $T(z)$ of conformal weight 2. The genus- g free energy $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT) receives no cross-channel correction: $\partial F_g^{\text{cross}} = 0$ for all $g \geq 1$.*

Proof. The cross-channel correction $\partial F_g^{\text{cross}}$ arises from the interaction between fields of different conformal weights in the propagator sum. The Virasoro algebra has a single generator of weight 2: there is only one channel, and the cross-channel correction vanishes identically. The scalar formula $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ is exact at all genera for this single-weight algebra. $\square$

7.5. The genus-1 fibre differential. The explicit genus-1 bar complex provides a direct computation of F_1 . The genus-1 propagator on the elliptic curve E_τ is $\eta^{(1)}(z_1, z_2; \tau) = \partial_{z_1} \log \mathfrak{P}_1(z_1 - z_2 | \tau) + 2\pi i \text{Im}(z_1 - z_2)/\text{Im} \tau$.

Proposition 7.7 (The three-term formula). *The fibre differential on the degree-2 bar element $[s^{-1}T | s^{-1}T]$ at genus 1 is:*

$$d_{\text{fib}}^{(1)}[s^{-1}T | s^{-1}T] = s^{-1} \left(:TT: - \frac{2\pi^2}{3} E_2(\tau) \cdot T - \frac{c\pi^4}{90} E_4(\tau) \cdot 1 \right),$$

where $E_2(\tau) = 1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n$ is the quasi-modular Eisenstein series and $E_4(\tau)$ is the weight-4 Eisenstein series. The three terms arise from:

- (i) $:TT:$: the genus-0 normal-ordered product (regular term paired with $1/u$);
- (ii) $-\frac{2\pi^2}{3} E_2(\tau) \cdot T$: the genus-1 state-dependent correction (double pole $2T/u^2$ paired with the propagator coefficient $-(\pi^2/3)E_2(\tau)u$);

(iii) $-\frac{c\pi^4}{90}E_4(\tau) \cdot \mathbf{1}$: the genus-1 vacuum correction (quartic pole $(c/2)/u^4$ paired with $-(\pi^4/45)E_4(\tau)u^3$).

Proof. The meromorphic part of the genus-1 propagator has the Laurent expansion $h(u; \tau) = 1/u - (\pi^2/3)E_2(\tau)u - (\pi^4/45)E_4(\tau)u^3 - \dots$ (odd powers only). The fibre differential is the residue of $h(u; \tau) \cdot T(z_1)T(z_2)$ at $u = 0$. The Virasoro OPE (??) has poles at orders 4, 2, and 1 in $u = z_1 - z_2$. Pairing the propagator terms $c_k(\tau)u^{2k-1}$ with the OPE terms $T_{(2k-1)}T u^{-(2k)}$ gives the residue: $T_{(-1)}T = :TT:$ from the leading $1/u$, $c_1 \cdot T_{(1)}T = -(\pi^2/3)E_2 \cdot 2T$ from $k = 1$, $c_2 \cdot T_{(3)}T = -(\pi^4/45)E_4 \cdot (c/2)$ from $k = 2$. $\square$

The E_2 -correction is quasi-modular (it transforms anomalously under $\tau \mapsto -1/\tau$); the E_4 -correction is fully modular. The Virasoro OPE pole structure generates precisely the Eisenstein tower $\{E_{2k}\}$ at genus 1.

8. BTZ BLACK HOLES AS MAURER–CARTAN ELEMENTS

The BTZ black hole of Bañados, Teitelboim, and Zanelli [BTZ92] is the unique black hole solution of 2+1-dimensional gravity with $\Lambda < 0$. We identify it as a Maurer–Cartan element of the gravitational convolution algebra of the algebraic sector. The identification is unconditional at the algebraic level (boundary $A_b = \text{Vir}_c$, MC element α_{BTZ}); the lift of α_{BTZ} to a saddle of a compact dynamical-metric path integral, and the recovery of microstate counts from a sum over geometries, requires the BTZ/ Cardy hypothesis package and the Maloney–Witten + Manschot–Moore Farey-tail completion (Section ??).

BTZ / Cardy hypothesis package (named for use throughout this section and Section ??):

- (BC1) *Modular invariance.* The torus partition function $Z_1(\tau)$ is invariant under $\text{SL}_2(\mathbb{Z})$, and in particular under the S -transform $\tau \mapsto -1/\tau$ used in the Cardy saddle-point.
- (BC2) *Vacuum dominance.* As $\tau \rightarrow i0^+$ along the imaginary axis, $Z_1(\tau)$ is dominated by the Virasoro vacuum representation: $Z_1(\tau) \sim e^{-2\pi i(-c/24)\tau} \cdot (1 + O(q))$.
- (BC3) *Saddle stability.* The saddle-point approximation to the modular S -transform is uncorrected at leading order in $\hbar \gg c$, with subleading corrections enumerated by the Hardy–Ramanujan circle method (Faulkner–Lewkowycz subleading; Manschot–Moore Farey expansion).
- (BC4) *Maloney–Witten Farey-tail completion.* The sum over $\text{SL}_2(\mathbb{Z})/\mathbb{Z}$ images of the BTZ saddle produces the Maloney–Witten partition function $Z_{\text{MW}}(\tau)$; the Manschot–Moore Farey-tail prescription regularises this (otherwise non-convergent / negatively-graded) sum into a holomorphic mock-modular completion.

The identification of α_{BTZ} as an MC element is unconditional. The Cardy entropy formula (Theorem ??) below uses (BC1)–(BC3); the genus- g saddle expansion (Proposition ??) uses (BC3); the construction of a compact gravitational partition function from the algebraic sector uses all four hypotheses and is the F12 open problem.

8.1. The BTZ solution. A BTZ black hole is specified by two parameters: mass M and angular momentum J , subject to $|J| \leq M\ell$. In the Chern–Simons formulation, the BTZ solution is a pair of flat $\mathfrak{sl}_2$ -connections with holonomy around the spatial circle:

Definition 8.1 (BTZ gauge configuration). The BTZ black hole with mass M and angular momentum J is the flat connection $A_{\pm}$ with holonomy eigenvalues

$$\exp(\pm 2\pi\sqrt{h}) \quad \text{and} \quad \exp(\pm 2\pi\sqrt{\bar{h}}),$$

where the conformal weights are

$$h = \frac{M\ell + J}{2}, \quad \bar{h} = \frac{M\ell - J}{2}.$$

The outer horizon radius is $r_+ = \ell\sqrt{8GM}(1 + \sqrt{1 - (J/(M\ell))^2})^{1/2}$.

8.2. BTZ as MC element.

Theorem 8.2 (BTZ as Maurer–Cartan element). *Type signature. Quadrant: Open. Presentation: chain-level (Maurer–Cartan in the modular convolution algebra of the algebraic sector). Level: 4 (operator-level: line / flat-connection encoding). Hypothesis package: $\Lambda < \circ(AdS_3)$; flat $A_{\pm}$ connection on the solid torus $D \times S^1$ with prescribed holonomy; identification of holonomy weights with conformal weights $(h, \bar{h})$ via Brown–Henneaux. The lift to a saddle of a compact dynamical-metric path integral requires $(BC1)$ – $(BC4)$ and is the F12 frontier.*

The BTZ black hole of mass M and angular momentum J is a Maurer–Cartan element

$$\alpha_{\text{BTZ}} = h e_h + \bar{h} e_{\bar{h}} \in \mathfrak{g}_{\text{Vir}_c}^{\text{mod}},$$

where $e_h, e_{\bar{h}}$ are the generators of the modular convolution algebra at conformal weights $h = (M\ell + J)/2$, $\bar{h} = (M\ell - J)/2$. The MC equation $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ is equivalent to the flatness condition $F_{\pm} = 0$ for the BTZ gauge connection.

Proof. The BTZ solution is a flat $SL(2, \mathbb{R})$ connection on the solid torus $D \times S^1$ with specified holonomy. In the boundary Virasoro description, flatness becomes the MC equation: the highest-weight state $|h, \bar{h}\rangle$ in the Virasoro module of weight $(h, \bar{h})$ satisfies the conformal Ward identities, which are exactly the components of the MC equation $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ projected to the boundary. $\square$

8.3. The Cardy entropy. The statistical entropy of the BTZ black hole is computed from the asymptotic growth of Virasoro characters.

Theorem 8.3 (Cardy formula and Bekenstein–Hawking entropy). *Type signature. Quadrant: Open. Presentation: chain-level + Cardy saddle. Level: 5 (modular trace of level-4 operators). Hypothesis package: BTZ / Cardy $(BC1)$ – $(BC3)$ above; trace direction (operators $\rightarrow$ scalar) only.*

The microstate entropy of the BTZ black hole at conformal weights $(h, \bar{h})$ with $h, \bar{h} \gg c$ is:

$$S_{\text{Cardy}}(h, \bar{h}) = 2\pi\sqrt{\frac{ch}{6}} + 2\pi\sqrt{\frac{c\bar{h}}{6}} = S_{\text{BH}},$$

where the Bekenstein–Hawking entropy is

$$S_{\text{BH}} = \frac{\pi r_+}{2G} = \frac{\text{Area}(\text{horizon})}{4G}.$$

Proof. The Cardy formula: the number of Virasoro primaries of weight h grows as $\rho(h) \sim \exp(2\pi\sqrt{ch/6})$ for $h \gg c$ (the saddle-point approximation to the modular S -transform of the partition function). For the BTZ state $(h, \bar{h})$: $S = \log \rho(h) + \log \rho(\bar{h}) = 2\pi\sqrt{ch/6} + 2\pi\sqrt{c\bar{h}/6}$.

The Bekenstein–Hawking entropy in terms of boundary data: using $c = 3\ell/(2G)$, $h = (M\ell + J)/2$, $\bar{h} = (M\ell - J)/2$:

$$\begin{aligned} S_{\text{Cardy}} &= 2\pi\sqrt{\frac{3\ell}{12G}(M\ell + J)} + 2\pi\sqrt{\frac{3\ell}{12G}(M\ell - J)} \\ &= 2\pi\sqrt{\frac{\ell}{4G} \cdot \frac{M\ell + J}{1}} + 2\pi\sqrt{\frac{\ell}{4G} \cdot \frac{M\ell - J}{1}} \\ &= \frac{\pi r_+}{2G}, \end{aligned}$$

where the last step uses the relation $r_{\pm}^2 = 8G\ell^2 M(1 \pm \sqrt{1 - (J/(M\ell))^2})/2$ for the non-rotating case $J = 0$: $r_+ = \ell\sqrt{8GM}$, $S_{\text{BH}} = \pi\ell\sqrt{8GM}/(2G) = \pi r_+/(2G)$. $\square$

8.4. Quantum corrections to BTZ entropy. The shadow obstruction tower provides a systematic genus expansion of quantum corrections to the leading Bekenstein–Hawking entropy.

Proposition 8.4 (BTZ genus expansion). *The entropy of the BTZ black hole admits a saddle-point expansion:*

$$S(M) = S_{\text{BH}} - \frac{3}{2} \log S_{\text{BH}} + \sum_{g=2}^{\infty} \frac{C_g(\kappa)}{S_{\text{BH}}^{2g-2}},$$

where $S_{\text{BH}} = 2\pi\sqrt{ch/6}$ and the coefficients $C_g(\kappa)$ depend on the shadow free energies $F_1, \dots, F_g$:

- (i) The leading correction is logarithmic: $-\frac{3}{2} \log S_{\text{BH}}$, universal across all $2+1$ -dimensional black holes.
- (ii) $F_1 = \kappa/24 = c/48$: the one-loop free energy from the scalar shadow.
- (iii) $F_2 = 7\kappa/5760 = 7c/11520$: the two-loop free energy.
- (iv) The shadow partition function $Z^{\text{sh}} = \exp(\sum_{g \geq 1} F_g \beta^{2g})$ converges absolutely (Bernoulli decay $\lambda_g^{\text{FP}} \sim (2\pi)^{-2g}$), in contrast to the string free energy which diverges as $(2g)!$.

Proof. The saddle-point expansion of $Z(\beta) = \exp(\sum_{g \geq 0} F_g \beta^{2g})$ around the BTZ saddle gives the $1/S_{\text{BH}}$ expansion. The logarithmic correction $-3/2$ is the universal contribution from the one-loop determinant of the graviton fluctuation around the BTZ geometry (the Jacobian of the saddle-point integral in three dimensions). The higher-order corrections C_g are determined by the shadow free energies through the saddle-point formula. $\square$

8.5. BTZ complementarity and the Hawking–Page transition.

Proposition 8.5 (BTZ paired entropy). *The Koszul pair $(\text{Vir}_c, \text{Vir}_{26-c})$ induces a paired entropy:*

$$S_{\text{BH}}(\text{Vir}_c; h) + S_{\text{BH}}(\text{Vir}_{26-c}; h) = 2\pi\sqrt{\frac{h}{6}}(\sqrt{c} + \sqrt{26-c}),$$

maximised at $c = 13$ by the concavity of $\sqrt{\cdot}$, where $\sqrt{c} + \sqrt{26-c}|_{c=13} = 2\sqrt{13}$. The Hawking–Page transition at $\beta_{\text{HP}} = 2\pi\ell$ exchanges the BTZ and thermal AdS saddles.

Proof. The paired entropy is $S_{\text{BH}}(\text{Vir}_c) + S_{\text{BH}}(\text{Vir}_{26-c}) = 2\pi\sqrt{ch/6} + 2\pi\sqrt{(26-c)h/6} = 2\pi\sqrt{h/6}(\sqrt{c} + \sqrt{26-c})$. By AM–QM inequality, $\sqrt{c} + \sqrt{26-c}$ is maximised when $c = 26 - c$, i.e. $c = 13$. The Hawking–Page temperature $\beta_{\text{HP}} = 2\pi\ell$ is the point where the BTZ free energy $F_{\text{BTZ}} = -(c/6\pi)(\pi^2/\beta)$ crosses the thermal AdS free energy $F_{\text{AdS}} = -(c/6\pi)\beta$. $\square$

Remark 8.6 (The entanglement entropy). The Calabrese–Cardy entanglement entropy of a boundary interval of length L is $S_{\text{EE}} = (c/3) \log(L/\varepsilon)$. The Koszul-dual entanglement entropy is $S'_{\text{EE}} = ((26 - c)/3) \log(L/\varepsilon)$. Their sum: $S_{\text{EE}} + S'_{\text{EE}} = (26/3) \log(L/\varepsilon)$, independent of c . At $c = 13$: $S_{\text{EE}} = S'_{\text{EE}} = (13/3) \log(L/\varepsilon)$.

9. THE ALGEBRAIC PAGE-CURVE SHADOW FROM COMPLEMENTARITY

Under the BTZ / Cardy hypothesis package (Section ??, (BC₁)–(BC₃)) and the island-saddle mechanism upstream (Penington [?Pen19], Almheiri–Engelhardt–Marolf–Maxfield [?AEMM19, ?AMM19]), the complementarity constant $K = \kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = 13$ controls the transition between the two Koszul-dual phases of the algebraic-sector saddle competition. The derivation below is the algebraic Page shadow of the gravitational Page curve: the Lagrangian decomposition of the genus expansion evaluated on the BTZ Maurer–Cartan element, the evaporation rate from the genus-1 free energy, and the Page time as the crossing of the two saddle-point contributions on the algebraic sector. The c -independent Page time $t_P = 3S_{\text{BH}}/13$ is a consequence of complementarity, not an assumption. The full information-theoretic Page curve requires the island mechanism on the dynamical-metric path integral (a distinct construction); the present theorem computes the algebraic-sector saddle exchange.

9.1. The Lagrangian decomposition on the BTZ background. The complementarity theorem (Theorem ??) decomposes the genus- g bar complex into Koszul-dual halves. We evaluate this decomposition on the BTZ Maurer–Cartan element $\alpha_{\text{BTZ}} \in \mathfrak{g}^{\text{mod}}$ (Theorem ??), producing a two-saddle partition function parametrised by the black hole mass.

Construction 9.1 (Lorentzian BTZ genus expansion). Let α_{BTZ} be the MC element at conformal weight h (non-rotating, $J = 0$). The genus-summed free energy (Proposition ??) is:

$$\mathcal{F}(h;) = S_{\text{BH}}(h) \quad^{-2} + \sum_{g \geq 1} F_g(\alpha_{\text{BTZ}}) \quad^{2g-2},$$

where $S_{\text{BH}}(h) = 2\pi\sqrt{ch}/6$ and $\quad = S_{\text{BH}}^{-1}$. The Lorentzian continuation replaces the Euclidean modular parameter $\tau = i\beta/(2\pi\ell)$ by $\tau \mapsto t/(2\pi\ell) + i\epsilon$, where t is Lorentzian time and $\epsilon \rightarrow 0^+$ selects the retarded contour. For an evaporating black hole, the conformal weight is time-dependent: $h(t) = h_0 - \dot{h} t$ with $\dot{h} > 0$, so that $S_{\text{BH}}(t) = 2\pi\sqrt{c h(t)}/6$ decreases monotonically.

Theorem 9.2 (Lagrangian decomposition on the BTZ background). *The complementarity decomposition (Theorem ??, Theorem C) applied to the genus expansion (??) produces:*

$$\mathcal{F}(h;) = \mathcal{F}_+(\kappa; h;) + \mathcal{F}_-(\kappa'; h;),$$

where the two Lagrangian contributions are:

$$\begin{aligned} \mathcal{F}_+(\kappa; h;) &= \kappa \quad^{-2} + \sum_{g \geq 1} \kappa \lambda_g^{\text{FP}} \quad^{2g-2}, & \kappa &= c/2, \\ \mathcal{F}_-(\kappa'; h;) &= \kappa' \quad^{-2} + \sum_{g \geq 1} \kappa' \lambda_g^{\text{FP}} \quad^{2g-2}, & \kappa' &= (26 - c)/2. \end{aligned}$$

The genus- g obstruction decomposes:

$$\mathbf{C}_g(\text{Vir}_c)|_{\alpha_{\text{BTZ}}} = \mathbf{Q}_g^+(\kappa) \oplus \mathbf{Q}_g^-(\kappa'),$$

with fiberwise differentials satisfying $(d_{\text{fib}}^+)^2 = \kappa \cdot \omega_g$ and $(d_{\text{fib}}^-)^2 = \kappa' \cdot \omega_g$ on the two summands.

Proof. Theorem C applied to the Virasoro algebra gives the Lagrangian eigenspace decomposition of the genus- g bar complex under the Koszul involution $c \mapsto 26 - c$. The involution has period 2; its two

eigenvalues on the fiberwise differential $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ are $\kappa = c/2$ and $\kappa' = (26 - c)/2$, the curvatures of the two eigenspaces. Since $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT, Proposition ??), each summand contributes proportionally to its curvature eigenvalue. The genus-summed partition function decomposes accordingly. The evaluation on α_{BTZ} preserves the decomposition: the MC equation $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ is satisfied independently on each Lagrangian half because the Koszul involution commutes with the MC operator (it acts on the parameter c , not on the MC variable α). $\square$

9.2. The radiation entropy from saddle-point competition. The two Lagrangian halves compete as the black hole evaporates. The radiation entropy is determined by the saddle-point prescription (the gravitational path integral selects the geometry minimising the generalised entropy), and the Page curve emerges from the exchange of dominance.

Definition 9.3 (Evaporation rate from the genus expansion). The entanglement production rate of a 2d CFT at central charge c and Hawking temperature T_H is determined by the genus-1 free energy. In thermal time units $\beta/(2\pi)$:

$$\dot{S}_{\text{rad}} = \frac{4\pi F_1}{\beta/(2\pi)} = 4\pi \cdot \frac{c}{48} = \frac{\pi c}{12} = \frac{c}{6} \text{ per unit } \frac{\beta}{2\pi},$$

using $F_1 = \kappa/24 = c/48$ (Theorem ??). The Koszul dual boundary, at central charge $26 - c$, has entanglement production rate $(26 - c)/6$.

Theorem 9.4 (The algebraic Page curve). *Type signature. Quadrant: Open. Presentation: scalar shadow on the algebraic sector (operator $\rightarrow$ scalar trace direction). Level: 5 (modular trace + saddle competition; scalar shadow output, not operator-level reconstruction). Hypothesis package: BTZ / Cardy (BC1)–(BC3); semiclassical $S_{\text{BH}} \gg 1$; Lagrangian decomposition under the Koszul involution (Theorem ??); island mechanism upstream (Penington, AEMM) supplies the information-theoretic content; the algebraic sector supplies the saddle-exchange shadow only.*

The radiation entropy of an evaporating BTZ black hole is

$$S_{\text{rad}}(t) = \min(S_+(t), S_{\text{BH}}(\text{o}) - S_-(t)),$$

where:

- $S_+(t) = (c/6)t$ is the no-island contribution (entanglement growth at the boundary rate $\dot{S}_+ = c/6$, controlled by $\kappa = c/2$);
- $S_-(t) = ((26 - c)/6)t$ is the island contribution (entanglement growth at the dual rate $\dot{S}_- = (26 - c)/6$, controlled by $\kappa' = (26 - c)/2$);
- $S_{\text{BH}}(\text{o})$ is the initial Bekenstein–Hawking entropy.

The following are consequences:

- (i) **Page time.** The crossing condition $S_+(t_P) = S_{\text{BH}}(\text{o}) - S_-(t_P)$ gives:

$$t_P = \frac{3 S_{\text{BH}}(\text{o})}{K} = \frac{3 S_{\text{BH}}(\text{o})}{13},$$

independent of c .

- (ii) **Page entropy.** $S_{\text{Page}} = S_+(t_P) = \frac{c S_{\text{BH}}(\text{o})}{26}$.

- (iii) **Hawking phase** ($t < t_P$): the boundary saddle $\bar{B}(\text{Vir}_c)$ with curvature $\kappa = c/2$ dominates; radiation entropy grows at rate $c/6$.

- (iv) **Island phase** ($t > t_P$): the Koszul dual saddle $\bar{B}(\text{Vir}_{26-c})$ with $\kappa' = (26 - c)/2$ dominates; radiation entropy decreases.

- (v) **Self-dual point** ($c = 13$): $\kappa = \kappa' = 13/2$, the two saddles have equal weight at all times, $S_{\text{Page}} = S_{\text{BH}}/2$.

Proof. The proof has three steps: the Lagrangian decomposition of the replica partition function, the identification of the radiation entropy from saddle dominance, and the computation of the crossing time.

Step 1: The replica partition function. The radiation entropy is computed from the replica partition function $Z(n)$ on the n -sheeted BTZ geometry: $S_{\text{rad}} = \lim_{n \rightarrow 1} (1 - n)^{-1} \log(Z(n)/Z(1)^n)$. The Lagrangian decomposition (Theorem ??) gives $Z(n) = Z_+(n) + Z_-(n)$, corresponding to the boundary and dual saddles. The n -sheeted replica of the BTZ geometry is itself a BTZ geometry with conical excess at the replica branch point; Theorem ?? identifies it as an MC element, so the genus expansion retains the same two-saddle decomposition at every replica number.

Step 2: The min formula from the generalised entropy. In the semiclassical regime $S_{\text{BH}} \gg 1$, the gravitational path integral is dominated by the saddle with smallest generalised entropy (the quantum extremal surface prescription). Two saddles contribute:

- The *no-island saddle*: the boundary-side partition function Z_+ , contributing $S_{\text{rad}} = S_+(t)$ to the radiation entropy. The rate $\dot{S}_+ = c/6$ is the genus-1 entanglement rate (Definition ??).
- The *island saddle*: the Koszul dual partition function Z_- , contributing $S_{\text{rad}} = S_{\text{BH}}(o) - S_-(t)$. The subtraction arises because the island saddle computes the complement: the radiation entropy equals the total entropy minus the interior entropy, and the interior is controlled by $\kappa' = (26 - c)/2$.

The dominant saddle is whichever gives the *smaller* radiation entropy: $S_{\text{rad}} = \min(S_+, S_{\text{BH}}(o) - S_-)$.

Step 3: The crossing time. The crossing condition $S_+(t_P) = S_{\text{BH}}(o) - S_-(t_P)$ reads:

$$\frac{c}{6} t_P = S_{\text{BH}}(o) - \frac{26 - c}{6} t_P.$$

Adding the two rates: $t_P(\frac{c}{6} + \frac{26 - c}{6}) = t_P \cdot \frac{26}{6} = S_{\text{BH}}(o)$. Solving:

$$t_P = \frac{6 S_{\text{BH}}(o)}{26} = \frac{3 S_{\text{BH}}(o)}{13} = \frac{3 S_{\text{BH}}(o)}{K}.$$

The c -independence is exact: the rates $c/6$ and $(26 - c)/6$ are individually c -dependent, but their sum $26/6 = \varkappa/3$ depends only on the scalar complementarity invariant $\varkappa := \kappa + \kappa' = 13$ (distinct from the Trinity ghost-charge conductor $K(\text{Vir}) := c + c^\dagger = 26$; related by $\varkappa = \rho_{\text{Vir}} \cdot K$, $\rho_{\text{Vir}} = 1/2$). $\square$

Remark 9.5 (The entanglement rate is not an external input). The Calabrese–Cardy entanglement rate $c/6$ is not imported from an independent argument. It is *derived* from the bar complex: the genus-1 free energy $F_1 = \kappa/24 = c/48$ (Theorem ??) determines the Hawking radiation rate via $\dot{S}_{\text{rad}} = 4\pi F_1/(\beta/(2\pi)) = c/6$ (Definition ??). The rate for the Koszul dual is $(26 - c)/6$, and their sum $26/6 = K/3$ is the quantity controlling the Page time. The bar complex provides both the two saddles and their relative growth rates; the entire Page curve is internal to the genus expansion.

Remark 9.6 (c -independence of the Page time). The coefficient $3/13$ in $t_P = 3S_{\text{BH}}/13$ is purely algebraic: 3 is the entanglement normalisation ($c/6$ rather than $c/2$; equivalently, $\kappa/3$ rather than κ), and 13 is the Virasoro complementarity constant $K = 13$. The dependence on the specific black hole enters only through S_{BH} . For any Koszul pair with complementarity constant K , the Page time is $t_P = 3S_{\text{BH}}/K$.

9.3. The island as Koszul dual dominance. The island of the quantum extremal surface prescription has a precise algebraic incarnation: it is the locus in the genus expansion where the Koszul dual saddle dominates. The complementarity theorem fixes the number of saddles (exactly two), their curvatures (κ and κ'), and the transition time (t_P).

Theorem 9.7 (Complementarity decomposition). *At each genus $g \geq 1$, the genus- g obstruction decomposes under the Koszul involution:*

$$\mathbf{C}_g(\text{Vir}_c) = \mathbf{Q}_g(\text{Vir}_c) \oplus \mathbf{Q}_g(\text{Vir}_{26-c}).$$

The curvature of the first summand is $\kappa = c/2$; the curvature of the second is $\kappa' = (26 - c)/2$. There is no third saddle: the Koszul involution has period 2, so the pair $(\text{Vir}_c, \text{Vir}_{26-c})$ exhausts the decomposition.

The transition is sharp: at $t = t_P$, the dominance switches at every genus simultaneously because the ratio $\kappa/\kappa' = c/(26 - c)$ is genus-independent (the λ_g^{FP} factors cancel in the ratio of Boltzmann weights).

Proof. The Lagrangian eigenspace decomposition is Theorem C applied to the Virasoro algebra (Theorem ??). The curvatures κ and κ' are the eigenvalues of $d_{\text{fib}}^2/\omega_g$ on the two summands. The ratio of Boltzmann weights at genus g is

$$\frac{Z_+^{(g)}}{Z_-^{(g)}} = \exp((\kappa - \kappa') \lambda_g^{\text{FP}} \cdot 2^{g-2}) = \exp((c - 13) \lambda_g^{\text{FP}} \cdot 2^{g-2}),$$

which depends on g through the common factor $\lambda_g^{\text{FP}} \cdot 2^{g-2}$ but whose sign ($c > 13$ vs $c < 13$) is genus-independent. At $c = 13$: the ratio is 1 at all genera and all times. $\square$

Remark 9.8 (Comparison with Penington and AEMM). The gravitational island of Penington [?Pen19] and Almheiri–Engelhardt–Marolf–Maxfield [?AEMM19, ?AMMZ19] is the locus where the quantum extremal surface dominates the gravitational path integral. The Koszul dual saddle $\bar{B}(\text{Vir}_{26-c})$ is the algebraic counterpart: the two-saddle structure, the exchange of dominance, and the c -independent Page time all emerge from the complementarity constant $K = 13$ alone. The identification:

QES / replica wormhole	Bar complex
no-island saddle	$\mathbf{Q}_g(\text{Vir}_c)$
island saddle	$\mathbf{Q}_g(\text{Vir}_{26-c})$
quantum extremal surface	Koszul involution fixed locus
island region	κ' -eigenspace dominance
Page time	saddle crossing
two saddles, no more	period-2 involution

The number of saddles (exactly two) is determined by the Koszul involution, not by a choice of replica geometry. The bar complex provides the mechanism; the Penington–AEMM programme provides the physical context and the fine-grained interpretation.

9.4. The transgression algebra and the Page transition. The transgression algebra B_Θ provides a second, independent characterisation of the Page transition. The degeneration of B_Θ at $u = \eta^2 = 0$ is the algebraic mechanism underlying the saddle exchange: the Morita-invisible Hawking phase degenerates into the information-carrying island phase.

Proposition 9.9 (The transgression algebra at the BTZ saddle). *Let B_Θ be the transgression algebra of the bar complex $\bar{B}(\text{Vir}_c)$ with MC element $\Theta = \Theta_{\text{Vir}_c}$ (Section ??). Set $u := \eta^2$ where η is the degree-one generator with $d\eta = \Theta$. At the BTZ saddle:*

- (i) *For $u \neq 0$ (the generic locus): the genus- g Clifford completion satisfies $\mathfrak{G}_g(B_\Theta)[u^{-1}] \cong \text{Mat}_{2^g}(B_\Theta[u^{-1}])$. Higher genus is Morita-invisible: the matrix algebra carries no new invariants beyond the genus-0 data. This is the **Hawking phase**: the boundary saddle dominates, and the black hole appears thermal.*
- (ii) *For $u = 0$ (the degenerate locus): $\mathfrak{G}_g(B_\Theta)/(\eta^2) \cong B/(\Theta) \otimes \Lambda^{2g+1}$, the exterior algebra degeneration. The $\binom{2g+1}{k}$ generators in Λ^k resolve the Morita-invisible matrix algebra into topological sectors*

of the genus- g partition function. This is the **island phase**: the Koszul dual saddle dominates, and the radiation carries fine-grained information.

- (iii) The **Page transition** is the passage from $u \neq 0$ to $u = 0$. The secondary anomaly $u = \eta^2$ vanishes when the two Lagrangian saddles reach equal weight. The locus $u = 0$ is the codimension-one wall in the space of MC deformations where the matrix algebra degenerates to the exterior algebra.

Proof. Parts (i) and (ii) are the two regimes of the Clifford completion. The Clifford algebra $\text{Cl}_{2g+1}(u)$ is the matrix algebra Mat_{2^g} for $u \neq 0$ (non-degenerate quadratic form) and the exterior algebra Λ^{2g+1} for $u = 0$ (zero form).

Part (iii): the BTZ MC element $\alpha_{\text{BTZ}} = h e_h$ determines the curvature $\Theta|_\alpha = \kappa \cdot \omega_1$ at genus 1. As the black hole evaporates, $h(t) \rightarrow 0$ and $\alpha_{\text{BTZ}}(t) \rightarrow 0$; correspondingly, $u(t) = \eta^2|_{\alpha(t)} \rightarrow 0$. The Page time t_P is the time at which $u(t_P) = 0$ to leading semiclassical order, which coincides with the saddle crossing (??) because $u = 0$ is the locus where neither Lagrangian half dominates. $\square$

Remark 9.10 (Morita invisibility is thermality). The Morita equivalence $\mathfrak{G}_g \cong \text{Mat}_{2^g}(B_\Theta[\eta^{-2}])$ for $u \neq 0$ is the algebraic expression of Hawking's argument: a thermal state carries no fine-grained information regardless of the number of replicas (g). The breakdown at $u = 0$ is the algebraic expression of unitarity restoration: the exterior algebra Λ^{2g+1} resolves the thermal degeneracy. The Page curve interpolates between these two regimes; the passage is controlled by the scalar complementarity $\varkappa := \kappa + \kappa' = 13$ (the Trinity $K(\text{Vir}) = 26$; $\varkappa = K/2$).

9.5. The Borel singularity and Stokes phenomenon. Conditionally on the genus-expansion model, the shadow free energies $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ converge (Bernoulli decay: $\lambda_g^{\text{FP}} \sim (2\pi)^{-2g}$). The full gravitational genus expansion, including the field-sector contributions from the infinite $\mathcal{A}_\infty$ tower (class **M**), diverges: the field-dependent terms $m_k|_T$ grow as $C^k \cdot k!$ (Gevrey-I, Remark ??). The Page transition has a third incarnation in this divergent structure: the Stokes phenomenon of the Borel-summed genus expansion.

Theorem 9.11 (Borel singularity from the Koszul dual). *The full gravitational genus expansion*

$$Z_{\text{grav}}(\) = \exp\left(\sum_{g \geq 0} F_g^{\text{full}} \cdot 2^{g-2}\right),$$

where F_g^{full} includes both the convergent scalar shadow $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT) and the Gevrey-I field-sector corrections, satisfies:

- (i) The series diverges factorially: $|F_g^{\text{full}}| \sim C^g \cdot (2g)!$, forced by class **M** (all $m_k \neq 0$).
(ii) The Borel transform $\hat{Z}(\zeta) = \sum_{g \geq 0} F_g^{\text{full}} \zeta^{2g-2} / (2g-2)!$ has a singularity at the instanton action

$$A_{\text{inst}} = \frac{|\kappa' - \kappa|}{K} \cdot (2\pi)^2 = \frac{|13 - c|}{13} \cdot 4\pi^2.$$

The singularity corresponds to the Koszul dual saddle $\bar{B}(\text{Vir}_{26-c})$.

- (iii) At $c = 13$ (self-dual): $A_{\text{inst}} = 0$. The Borel singularity sits at the origin, the two saddles have equal action, and the Stokes phenomenon is maximally strong.
(iv) The Stokes wall of the Borel summation is the Page time: the exponentially suppressed correction $\sim e^{-A_{\text{inst}}/}$ becomes of order unity at $t = t_P$ and dominates thereafter.

Proof. Part (i): the shadow depth $r_{\text{max}} = \infty$ (all $m_k \neq 0$, Theorem ??) forces the field-sector contributions to grow as $\|m_k|_T\| \sim C^k \cdot k!$ by the Virasoro OPE recursion. The cumulative genus- g free energy includes the sum over stable graph topologies at genus g with vertices decorated by $\{m_k\}$; the factorial growth of m_k combined with the graph-counting factor produces $|F_g^{\text{full}}| \sim C^g \cdot (2g)!$ (Gevrey-I). The

scalar shadow $\kappa \cdot \lambda_g^{\text{FP}} \sim \kappa \cdot (2\pi)^{-2g}$ converges (Bernoulli decay) but the field sector dominates for $g \gg 1$.

Part (ii): in the resurgence framework, the Borel singularity nearest to the origin corresponds to the non-perturbative saddle closest in action. Complementarity provides exactly one such saddle: $\tilde{B}(\text{Vir}_{26-c})$ with curvature κ' . The tree-level free energies are $\mathcal{F}_+^{(0)} = \kappa^{-2}$ and $\mathcal{F}_-^{(0)} = \kappa'^{-2}$; the action difference is $|\mathcal{F}_+^{(0)} - \mathcal{F}_-^{(0)}| = |\kappa - \kappa'|^{-2} = |c - 13|^{-2}/1$. Normalising by the scalar instanton action $(2\pi)^2$ (the leading pole of the $\hat{A}$ -genus at $x = 2\pi i$) gives $A_{\text{inst}} = (2\pi)^2 |\kappa' - \kappa|/K$.

Part (iii): at $c = 13$, $\kappa = \kappa'$, so $A_{\text{inst}} = 0$.

Part (iv): the Stokes wall is crossed when the Borel contour passes through the singularity. On one side: $\tilde{B}(\text{Vir}_c)$ dominates. On the other: $\tilde{B}(\text{Vir}_{26-c})$ dominates. The Stokes multiplier is ± 1 (the involution has period 2). The wall is crossed when the evaporating black hole reaches the critical value $t_P^{-1} = S_{\text{BH}}(t_P)$; at leading order this reproduces $t_P = 3S_0/13$. $\square$

Remark 9.12 (Three incarnations of the Page transition). The Page transition has three equivalent descriptions, each originating from a different structure on the bar complex:

- (1) **Lagrangian**: the crossing of the two Koszul-dual saddle-point contributions to the gravitational partition function (Theorem ??).
- (2) **Transgression**: the degeneration $u = \eta^2 \rightarrow 0$ in the transgression algebra B_Θ , where the Morita-invisible matrix algebra degenerates to the exterior algebra (Proposition ??).
- (3) **Resurgent**: the Stokes wall of the Borel-summed genus expansion, where the exponentially suppressed Koszul dual contribution becomes dominant (Theorem ??).

All three are controlled by the single datum $\varkappa := \kappa + \kappa' = 13$ (the scalar complementarity; the Trinity $K(\text{Vir}) = c + c' = 26$ sits upstream through $\varkappa = \varrho_{\text{Vir}} \cdot K = K/2$): it determines the crossing time, the degeneration locus, and the instanton action. The number of saddles is fixed by complementarity (period-2 involution), the degeneration type by the Clifford structure (matrix vs. exterior), and the Stokes multiplier is ± 1 (the involution sign).

10. DE SITTER ENTROPY AND THE GRAVITATIONAL YANGIAN

10.1. Analytic continuation to de Sitter. The algebraic Page curve exploits the Koszul pair $(\text{Vir}_c, \text{Vir}_{26-c})$ at fixed signature. Analytic continuation to Lorentzian de Sitter signature ($\ell \mapsto i\ell$, equivalently $\Lambda = -1/\ell^2 \mapsto +1/\ell^2$) preserves the shadow obstruction tower and the complementarity structure.

Theorem 10.1 (De Sitter shadow tower). *Under the Wick rotation $\ell \mapsto i\ell$ mapping $\text{AdS}_3 \rightarrow \text{dS}_3$:*

- (i) *The central charge continues to $c_{\text{dS}} = 3\ell_{\text{dS}}/(2G)$, and the curvature to $\kappa_{\text{dS}} = c_{\text{dS}}/2$.*
- (ii) *The Gibbons–Hawking entropy of the cosmological horizon is:*

$$S_{\text{dS}} = \frac{\pi c_{\text{dS}}}{3} = \frac{\pi \ell_{\text{dS}}}{2G} = \frac{\text{Area}(\text{cosmological horizon})}{4G}.$$

- (iii) *The shadow obstruction tower continues with real quantum corrections:*

$$F_g^{\text{dS}} = \kappa_{\text{dS}} \cdot \lambda_g^{\text{FP}} = \frac{c_{\text{dS}}}{2} \cdot \lambda_g^{\text{FP}} \quad (\text{UNIFORM-WEIGHT}).$$

In particular: $F_1^{\text{dS}} = c_{\text{dS}}/48$.

- (iv) *The Nariai limit $c_{\text{dS}} = 13$ is the analytic continuation of the self-dual point: the de Sitter cosmological horizon entropy at the Nariai bound is self-dual under $c_{\text{dS}} \mapsto 26 - c_{\text{dS}}$.*

Proof. The Brown–Henneaux formula $c = 3\ell/(2G)$ is algebraic in ℓ and continues under $\ell \mapsto i\ell$ (or, equivalently, $\ell_{\text{dS}} = |\ell|$). The curvature $\kappa = c/2$ is polynomial in c and continues directly. The Gibbons–Hawking entropy: the cosmological horizon of dS_3 has circumference $2\pi\ell_{\text{dS}}$, so $\text{Area} = 2\pi\ell_{\text{dS}}$, and $S_{\text{dS}} = 2\pi\ell_{\text{dS}}/(4G) = \pi\ell_{\text{dS}}/(2G) = \pi c_{\text{dS}}/3$. The shadow free energies $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ are polynomial in κ and continue to real values. The Nariai limit corresponds to the maximal black hole in dS_3 , where the black hole and cosmological horizons coincide; the central charge at this point is $c_{\text{dS}} = 13$, the self-dual value. $\square$

Remark 10.2 (The Banks conjecture). The Banks conjecture states that the Hilbert space of de Sitter quantum gravity is finite-dimensional: $\dim(\mathcal{H}_{\text{dS}}) = \exp(S_{\text{dS}}) = \exp(\pi c_{\text{dS}}/3)$. The convergent shadow obstruction tower (Bernoulli decay: $\lambda_g^{\text{FP}} \sim (2\pi)^{-2g}$) is consistent with this finite-dimensionality: the quantum corrections to the de Sitter entropy do not produce a divergent series that would require non-perturbative completion from additional degrees of freedom.

10.2. Horizon thermodynamics from negative curvature. The analytic continuation $\kappa \rightarrow -|\kappa|$ has a direct interpretation in the bar complex.

Proposition 10.3 (Negative curvature and horizon thermodynamics). *The fiberwise differential at genus $g \geq 1$ satisfies $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ on $\overline{\mathcal{M}}_g$. Under $\kappa \rightarrow -|\kappa|$:*

- (i) *The curved bar complex acquires opposite curvature: $d_{\text{fib}}^2 = -|\kappa| \cdot \omega_g$.*
- (ii) *The genus tower changes sign: $F_g^{\text{dS}} = -|\kappa| \cdot \lambda_g^{\text{FP}} < 0$ for $g \geq 1$.*
- (iii) *The negative free energies produce positive entropy contributions: $-F_g < 0$ in the exponent of Z means the partition function decreases, consistent with a finite-dimensional Hilbert space.*

The de Sitter entropy $S_{\text{dS}} = \pi c_{\text{dS}}/3$ is the genus-0 contribution (the classical action evaluated on the cosmological horizon saddle), with the tower of negative corrections $\{F_g^{\text{dS}}\}_{g \geq 1}$ systematically reducing the effective number of microstates.

Proof. The fiberwise differential is $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ (proved in Section ??). Replacing κ by $-|\kappa|$ changes the sign of ω_g . The genus- g free energy $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ then becomes $F_g^{\text{dS}} = -|\kappa| \cdot \lambda_g^{\text{FP}} < 0$ (since $\lambda_g^{\text{FP}} > 0$). $\square$

10.3. The gravitational Yangian at $c = 13$. At the Koszul self-dual point $c = 13$, the Virasoro algebra acquires a distinguished algebraic structure: the ordered bar complex $\bar{B}^{\text{ord}}(\text{Vir}_{13})$ carries a gravitational Yangian symmetry.

Definition 10.4 (Gravitational Yangian). The *gravitational Yangian* $\mathcal{Y}_{\text{grav}}$ is the algebra of conserved charges of the shadow obstruction tower $\{S_r\}_{r \geq 2}$ at $c = 13$, acting on the ordered bar complex $\bar{B}^{\text{ord}}(\text{Vir}_{13}) = T^c(S^{-1}\overline{\text{Vir}}_{13})$. The shadow connection $\nabla^{\text{hol}} = d - \sum_{r \geq 2} S_r$ at $c = 13$ is the generating function of the Yangian generators.

Theorem 10.5 (Properties of the gravitational Yangian at $c = 13$). *At the self-dual point $c = 13$:*

- (i) **Self-duality.** $\text{Vir}_{13}^! = \text{Vir}_{13}$: *the boundary algebra equals its Koszul dual. The curvature is $\kappa = 13/2$, and $\kappa_{\text{eff}} = (13 - 26)/2 = -13/2$.*
- (ii) **Symmetric genus tower.** $F_g(13) = F_g(26 - 13) = F_g(13)$: *the genus tower is self-dual. The free energies: $F_1 = 13/48$, $F_2 = 91/11520$.*
- (iii) **Symmetric Page curve.** At $c = 13$: $\kappa = \kappa' = 13/2$, *the two saddles have equal action, the Page curve is symmetric about t_P , and $S_{\text{Page}} = S_{\text{BH}}/2$.*

(iv) **Infinite conserved charges.** The shadow obstructions S_r , $r \geq 2$, at $c = 13$ satisfy:

$$\begin{aligned} S_2 &= 13/2, & S_3 &= -13, & S_4 &= 10/(13 \cdot 87) = 10/1131, \\ S_5 &= -48/(169 \cdot 87) = -48/14703. \end{aligned}$$

The discriminant $\Delta = 8 \cdot (13/2) \cdot 10/1131 = 40/87 \neq 0$ confirms class **M**: the gravitational Yangian has infinitely many generators.

(v) **R-matrix at self-duality.** The classical r -matrix at $c = 13$ is $r_{13}(z) = (13/2)/z^3 + 2T/z$, and the CYBE holds by Proposition ???. The KZ connection (??) at $c = 13$ has self-dual monodromy: the braiding σ satisfies $\sigma^2 = \sigma_{\text{dual}}^2$ (the braiding of Vir_{13} and $\text{Vir}_{13}^!$ coincide).

Proof. Parts (i)–(iii) are immediate from Theorem ?? and Proposition ?? at $c = 13$.

Part (iv): the shadow invariants at $c = 13$ are obtained by evaluating the general formulas: $S_2 = 13/2$, $S_3 = -13$, $S_4 = 10/(13(5 \cdot 13 + 22)) = 10/(13 \cdot 87) = 10/1131$, $S_5 = -48/(13^2 \cdot 87) = -48/14703$. The discriminant $\Delta = 8\kappa S_4 = 8(13/2)(10/1131) = 40/87 \neq 0$; class **M**.

Part (v): the r -matrix at $c = 13$ is $r_{13}(z) = (13/2)/z^3 + 2T/z$ by direct evaluation of (??). The CYBE holds for all c by Proposition ???. Koszul self-duality of the monodromy at $c = 13$ (i.e. $\text{Vir}_{13}^! = \text{Vir}_{13}$), the Koszul involution $c \mapsto 26 - c$ fixing $c = 13$) implies that the Koszul dual connection is the same connection. $\square$

Remark 10.6 (The role of $c = 13$ versus $c = 26$). The two distinguished central charges are:

- $c = 13$: *Koszul self-duality.* $\text{Vir}_{13}^! = \text{Vir}_{13}$. The complementarity constant is $K = 13$. The algebraic Page curve is symmetric. The gravitational Yangian has full self-dual structure. This is the algebraic self-duality of 3d gravity.
- $c = 26$: *matter-ghost cancellation.* $\kappa_{\text{eff}} = 0$. The effective genus tower vanishes: $F_g^{\text{eff}} = 0$ for all $g \geq 1$. The gravitational partition function becomes S -invariant. This is the bosonic string critical point, not self-duality.

The physical content of 3d gravity concentrates at $c = 13$, not $c = 26$: at $c = 13$, the gravitational perturbation series (the shadow obstruction tower) has maximal symmetry (self-duality), the algebraic Page curve has maximal symmetry (equal entropies), and the Yangian has maximal structure (self-dual braiding). At $c = 26$, the effective perturbation series vanishes.

Conjecture 10.7 (The gravitational Yangian as symmetry algebra). *At $c = 13$, the gravitational Yangian $\mathcal{Y}_{\text{grav}}$ is the symmetry algebra of 3d self-dual quantum gravity in the following sense:*

- The shadow obstructions $\{S_r\}_{r \geq 2}$ generate an algebra isomorphic to a quotient of the Yangian $Y(\mathfrak{vir}_{13})$ associated to the Virasoro Lie algebra at $c = 13$.
- The shadow connection $\nabla^{\text{hol}} = d - \sum_{r \geq 2} S_r$ is a connection for $\mathcal{Y}_{\text{grav}}$ -modules.
- The integrability of the self-dual gravitational theory (manifested by the convergence of the scalar shadow tower) is a consequence of the Yangian symmetry.
- The conserved charges S_r at $c = 13$ satisfy the RTT-type commutation relations

$$R_{12}(z - w) T_1(z) T_2(w) = T_2(w) T_1(z) R_{12}(z - w)$$

with R -matrix constructed from $r_{13}(z)$.

Remark 10.8 (Comparison with the holographic datum). The gravitational Yangian at $c = 13$ is the most structured specialisation of the holographic datum $\mathcal{H}(\text{Vir}_c)$: at $c = 13$, all seven entries are Koszul self-dual (invariant under $c \mapsto 26 - c$), the algebraic Page curve is symmetric, the genus tower has maximal symmetry, and the Yangian structure organises the infinite $\mathcal{A}_\infty$ tower into a representation-theoretic

framework. The holographic datum at generic c carries the same seven entries but without the Koszul self-duality (specific to $c = 13$) that enables the Yangian interpretation.

II. GRAVITY, TOPOLOGIZATION, AND CELESTIAL HOLOGRAPHY

The algebraic-sector identification of 3d gravity is not a motivational appendix to the modular Koszul programme. The algebraic sector is the first nontrivial gravitational realisation of the homotopy-completed iterated Sugawara topologisation ladder (conditional, scope-tagged), the chiral higher-Deligne identification of the bulk, the algebraic shadow of celestial holography, and the topologization tower across the four shadow classes. None of these constructions builds the compact dynamical-metric path integral; that lift is the F12 frontier (Section ??).

Theorem II.1 (Conditional iterated Sugawara topologisation ladder). *For a chiral algebra $\mathcal{A}$ with a tower of inner higher-spin Casimirs $\{T^{(n)}\}_{2 \leq n \leq N+1}$ each admitting a BRST primitive $G^{(n)}$ with $T^{(n)} = [Q_{\text{tot}}, G^{(n)}]$ on Q -cohomology, assume in addition that the primitives are BRST-commuting up to coherent homotopy, $[G^{(n)}, G^{(m)}] = Q$ -exact, and determine independent topological directions. Then the transferred derived chiral centre carries an E_{N+2} -topological structure via iterated Dunn additivity. Specialisations:*

- $\mathcal{A} = \text{Vir}_c$: one inner stress tensor, cohomological and weight-completed E_3^{top} ;
- $\mathcal{A} = \mathcal{W}_N$: inner tower of depth $N - 1$, E_{N+1}^{top} under the same coherent-homotopy package;
- $\mathcal{A} = \mathcal{W}_\infty$: infinite depth, E_∞^{top} only under convergence and completion hypotheses.

Consequently, 3d quantum gravity is modeled by the $N = 2$ shadow of this conditional topological tower.

Theorem II.2 (Chiral higher Deligne and holographic shadow). *The derived chiral centre $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c)$ is acted on, in the cohomological and weight-completed sense, by $\text{SC}^{\text{ch}, \text{top}}$ through the heptagon edges (3) $\leftrightarrow$ (4) $\leftrightarrow$ (5) of the Vol II $\text{SC}^{\text{ch}, \text{top}}$ heptagon. On the C_2 -cofinite non-logarithmic standard landscape, this gives the algebraic shadow of celestial holography: the celestial chiral algebra $\mathcal{A}^{\text{cel}}$ of boundary soft modes, the $\text{SC}^{\text{ch}, \text{top}}$ structure on $(\mathcal{A}^{\text{cel}}, Z_{\text{ch}}^{\text{der}}(\mathcal{A}^{\text{cel}}))$, and the shadow-tower coefficients reproducing the soft-factor hierarchy. Coverage: self-dual gauge (KM), gauge+matter (DS), gravity (Virasoro + $w_{1+\infty}$).*

Remark II.3 (Topologization tower: scope per class). Topologization of $Z_{\text{ch}}^{\text{der}}(\mathcal{A})$ has four regimes, with scope tags attached for every reference in this paper:

- Class G (Heisenberg): E_3^{top} immediate via Dunn on F^{H_k} (commutative $\Rightarrow$ Dunn applies).
- Class L ($V_k(\mathfrak{g})$, $k \neq -h^\vee$): E_3^{top} on the *original* complex via Costello–Gwilliam factorization on $\mathbb{R} \times \mathbb{C}$ with explicit topologizing primitives $\eta_i^{(i)}$, $\eta_i^{(ii)}$ making $[Q, \tilde{G}_1] = T_{\text{Sug}}$ strict chain-level.
- Class C ($\beta\gamma$, bc): E_3^{top} via Friedan–Martinec–Shenker bosonisation, reducing to Heisenberg.
- Class M (Vir , $\mathcal{W}_N$): E_3^{top} weight-completed via half-BRST + MC5 pattern; original-complex chain-level *open* (sole remaining direction of the topologization tower theorem).
- Critical level $k = -h^\vee$: E_2^{top} (dimension drop, not failure).

Theorem ?? is the class M line of this tower, attached with the cohomological / weight-completed hypothesis package.

Remark II.4 (Class M scope and logarithmic exceptions). Class M chain-level topologization on the *original* complex remains open; the weight-completed statement is proved. The scope qualifications are: (i) $W(p)$ triplet tempering is conjectural pending an Adamović–Milas character-amplitude bound (Gurarie 1993 + Flohr 1996 counterexamples); (ii) non-tempered stratum is *empty* only on the non-logarithmic subset; (iii) the conditional harmonic β_N law separates the two polynomial candidates at

$N = 4$ (13 versus 15 and 16), a parameter-question not structural; (iv) super-Yangian canonical pairing is sub-Sugawara. 3d quantum gravity with $\Lambda < 0$, the Virasoro climax at $c = 3\ell/(2G)$, and the c -linearity/self-duality structure are unconditional on the Virasoro Koszul locus (generic c); the class M chain-level conditional affects only original-complex chain-level bar/BV comparison, not the genus expansion computed in Section ??.

Remark 11.5 (Algebraic identification, not gravitational construction). The relationship between the Virasoro Beilinson tower and 3d gravity is an *identification of the algebraic sector*, not a construction of a compact dynamical-metric path integral. The universal Maurer–Cartan element Θ_{Vir_c} of the modular convolution algebra encodes, via its MC projections to successive degrees, the complete *algebraic-sector trace* of the perturbative gravitational genus expansion. Koszul duality $\text{Vir}_c \leftrightarrow \text{Vir}_{26-c}$ is the algebraic shadow of S-duality; Page-time complementarity $t_P = 3S_{\text{BH}}/13$ is the structural shadow of the Page transition. The full information-theoretic Page curve (Penington, AEMM) requires the island mechanism on the dynamical-metric path integral; the complementarity decomposition computes the saddle-exchange shadow on the algebraic sector. BTZ black holes are MC elements of the algebraic convolution algebra; de Sitter entropy emerges from analytic continuation $\kappa \rightarrow -|\kappa|$.

The trace direction is unconditional on the Virasoro Koszul locus. The inverse direction (algebraic sector $\rightarrow$ compact dynamical-metric path integral) is open frontier F12: it requires the BTZ / Cardy hypothesis package (modular invariance + vacuum dominance + saddle stability) plus the Maloney–Witten + Manschot–Moore Farey-tail completion, and it remains the named operator-level Pfaffian / square-root reconstruction problem (Vol I F11 + F12). The iterated Sugawara ladder (Theorem ??) embeds the present algebraic sector as the $N = 2$ truncation of a conditional E_∞ topologisation tower; the unconditional content is the $N = 2$ algebraic identification proved here.

12. THE F12 FRONTIER: FROM ALGEBRAIC SECTOR TO COMPACT DYNAMICAL-METRIC PATH INTEGRAL

The preceding sections constructed the algebraic holographic HT sector of 3d gravity at $\Lambda < 0$: the open Beilinson tower with boundary Vir_c , bulk $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) = C_{\text{ch}}^\bullet(\text{Vir}_c, \text{Vir}_c)$, and the $\text{SC}^{\text{ch}, \text{top}}$ -brace structure on the pair. The trace direction (operators $\rightarrow$ scalar shadow) is unconditional on the Virasoro Koszul locus: the genus tower $F_g = \kappa \lambda_g^{\text{FP}}$, the algebraic Page shadow $t_P = 3S_{\text{BH}}/13$, the Cardy / Bekenstein–Hawking entropy, the de Sitter analytic continuation, and the gravitational Yangian at $c = 13$ are all level-5 scalar shadows of the level-4 operator algebra, computed via modular trace and clutching on the open category.

The *inverse direction*, namely the reconstruction of a compact dynamical-metric path integral from the algebraic sector, is the open frontier F12. Stating this precisely:

Conjecture 12.1 (F12: algebraic sector $\rightarrow$ dynamical-metric path integral). *Type signature. Quadrant: Open. Presentation: operator-level Pfaffian / square-root reconstruction. Level: 5 $\rightarrow$ 4 (inverse modular reconstruction; scalar-shadow to operator-algebra comparison). Hypothesis package: BTZ / Cardy (BC1)–(BC4) of Section ??; Maloney–Witten partition function $Z_{\text{MW}}(\tau)$ defined via the $\text{SL}_2(\mathbb{Z})/\mathbb{Z}$ Farey sum over BTZ images; Manschot–Moore mock-modular Farey-tail completion; Hall–Borcherds residual + virtual K_0 -determinant package (Vol I F11).*

There exists a compact holomorphic-topological saddle structure on the gravitational path integral

$$Z_{3\text{dQG}, \text{Vir}}(S^2 \times S_\tau^1) = q^{(1-c)/24} (1-q)/\eta(\tau),$$

extending the algebraic-sector partition function of the present construction, such that:

- (i) $Z_{3\text{dQG}, \text{Vir}}$ equals the Maloney–Witten Farey-tail-completed sum over BTZ saddles plus thermal AdS_3 ;

- (ii) *the inverse trace map sends the level-5 scalar tower $\{S_r\}_{r \geq 2}$ to the level-4 operator content of the BTZ ensemble;*
- (iii) *the algebraic Page shadow $t_P = {}_3S_{\text{BH}}/13$ lifts to the full information-theoretic Page curve of Penington–AEMM via the island mechanism applied to the dynamical-metric path integral.*

Remark 12.2 (F12 comparison problem). F12 is the conjunction of the following three constructions:

- (1) *Operator-level Pfaffian* for the Virasoro family, analogous to the Igusa cusp form Δ_5 on $K_3 \times E$ (Vol I F11). The construction would identify the Manschot–Moore Farey-tail completion as a virtual K_0 -determinant of an explicit operator-level object whose protected Pfaffian is the Maloney–Witten partition function.
- (2) *Cross-volume bridge to Vol II / III*. Vol II’s $\text{SC}^{\text{ch}, \text{top}}$ -brace bulk is the operator-level companion of the algebraic sector identified here; the dynamical-metric path integral would emerge as the gravity-line completion of the Vol II climax. Vol III’s Drinfeld double $G(X) = D(Y^+(X))$ supplies the line / brane operator data.
- (3) *Resolution of the Maloney–Witten negativity problem*. The naive Maloney–Witten sum has negative Fourier coefficients in the partition function; Manschot–Moore Farey-tail completion is the candidate cure, but its compatibility with the Beilinson-tower reconstruction at level 4 is unverified. The resolution is the F12 endpoint.

Until F12 is closed, every statement of the form “the algebraic sector *is* the gravitational path integral” or “the modular form $Z_{3\text{d QG}}$ *constructs* the BPS Hilbert space” has the wrong type: scalar shadow $\neq$ operator algebra. The trace direction is unconditional; the inverse direction remains the open F12 comparison problem with its own hypothesis package.

Remark 12.3 (The cross-volume vertical). The algebraic holographic HT sector identified here sits at levels 3–5 of the Vol I Beilinson tower:

- Level 1 (chart): $A_b = \text{Vir}_c$.
- Level 2 (twisting): $\bar{B}^{\text{ord}}(\text{Vir}_c)$.
- Level 3 (bulk): $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) = C_{\text{ch}}^\bullet(\text{Vir}_c, \text{Vir}_c)$, with E_3^{top} structure (cohomological + weight-completed scope; Class **M**).
- Level 4 (operator): BTZ MC elements, Wilson lines, Vir_{26-c} -modules.
- Level 5 (scalar shadow): genus tower $F_g = \kappa \lambda_g^{\text{FP}}$, Cardy entropy, algebraic Page shadow.

Vertical equivalences: level 3 of Vol I corresponds to the Vol II $\text{SC}^{\text{ch}, \text{top}}$ -brace bulk (climax of Vol II); level 5 corresponds to the gravity-line completion (Vol II) and to the Borchers-weight identity $\kappa_{\text{BKM}}(\Phi_N) = c_N(o)/2$ (Vol III). F12 is precisely the vertical equivalence at level 4 in the gravitational direction: identifying the BTZ ensemble of the algebraic sector with the saddle ensemble of the compact dynamical-metric path integral.

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